Transportation Research Part C 191 (2026) 105768



Contents lists available at ScienceDirect

Transportation Research Part C

journal homepage: www.elsevier.com/locate/trc





Predictive stochastic repurposable supply chain: an adaptive strategy toward supply chain plasticity

Hani Gilani *, Tom Van Woensel

Department of Industrial Engineering and Innovation Sciences, Eindhoven University of Technology, Eindhoven 5600MB, Netherlands

ARTICLE INFO

Keywords:
Supply chain management
Disruption risk management
Stochastic programming
Artificial intelligence
Benders decomposition

ABSTRACT

The supply chain resilience concept traditionally implies disruptions that occur on a predictable scale and with specific durations, and assumes recovery activities will begin when the disruption ends. However, long-term crises such as trade wars and pandemics are generally unpredictable, persistent over long periods, and require that their recovery activities be begun during the crisis itself. In this case, returning to the old normal is often unrealistic and infeasible; thus, the supply chain must adapt to the new normal to maintain survivability and long-term viability. The present study introduces a new concept in supply chain management, the Repurposable Supply Chain (RSC), in which the supply chain adapts to long-term crises by shifting its operations to emerging product networks. In this regard, a two-stage stochastic MILP model is developed under uncertain disruption severity over the planning horizon. Since scenario generation is a complex process in such long-term crises, a predictive algorithm, equipped with machine-learning models that use real-time crisis-severity data, prunes the scenario tree and estimates the probabilities of the remaining scenarios. To solve the proposed RSC stochastic optimization problem at a large scale, a Generalized Benders Decomposition approach is developed and implemented. By implementing the proposed predictive algorithm, the number of potential scenarios is reduced from 729 to 8, which shows substantial pruning of the scenario tree. Income analysis on the case study reveals that the RSC performed at a higher level of the non-repurposable model, resulting in a 9.7% increase in total income, which relevant sensitive analysis proves the robustness of the result. Purchasing cost has been identified as the most sensitive term, with a 20% increase raising the objective value by 7.5% in the RSC compared to 8.8% in the non-RSC in the proposed case study. Stochastic analysis reveals that the learning-powered RSC more effectively handles uncertainty and captures its impacts. Finally, the proposed solution method effectively handles large-scale RSC instances within reasonable time limits and shows significant computational performance.

1. Introduction

Resilient Supply Chain Management (SCM) refers to managing the supply chain during a disruptive event; in other words, it reflects the supply chain's ability to absorb shocks and recover from severe disruptions, returning to the 'old normal' state (Fiksel, 2006, Müllerklein and Fontaine, 2025). Most of the literature on this concept considers the occurrence of disruptive events with specific characteristics, such as predictable scale and a known, specific duration, and recovery efforts begin when the disruption ends (Hosseini

* Corresponding author.
E-mail addresses: h.gilani@tue.nl, hani_gilani@ind.iust.ac.ir (H. Gilani).

https://doi.org/10.1016/j.trc.2026.105768
Received 29 December 2025; Received in revised form 16 May 2026; Accepted 18 May 2026
Available online 18 June 2026
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et al., 2019, Gilani et al., 2025b). However, in many cases (e.g., a trade war or a pandemic), these characteristics might change significantly. In the real world, some disruptions have specific characteristics: they are unpredictable in scale, last for an unknown period, and the recovery process can be conducted for them even while the disruption is still ongoing. The events with these characters are known as super-disruption in SCM (Ivanov, 2020).

Such events, compared to the traditional ones, behave differently in the supply chain (SC) environment, so they need to be considered in a reshaped supply chain, especially in the market. Decision-making during such events often extends beyond the traditional scope of the SC, as evidenced in the SC literature during the COVID-19 pandemic (Ivanov, 2018). The nature of these events creates a significant difference in conducting the recovery process, where a return to 'old normal' is no longer possible. In contrast, the supply chain prioritizes its survivability by adapting and evolving with what is named 'new normal.' This adoptive behavior has revolutionized the resilience SC in a new framework termed the 'viable supply chain.' A viable system frequently adapts to internal and external changes to survive long-term crises. (Ivanov and Dolgui, 2020).

As mentioned, adaptation plays a critical role in dealing with super-disruptions in SC, where it, by enabling specific mechanisms, helps monitor, anticipate, and adjust to an unstable environment during a long-term crisis (Ivanov, 2018). In particular, the SC organization is enabled to survive and remain viable under such disruptions through adoption mechanisms generally implemented through a reconfiguration. A novel and effective adoption strategy is to use product-emerging supply chains during the disruption itself, which often appears as a situation necessity. This insight introduces a new concept in supply chain management that we named 'Repurposable Supply Chain,' as a novel type of adoption in SC resilience.

The COVID-19 pandemic was a significant situation in which the potential of the term Repurposable Supply Chain (RSC) emerged for the first time. For example, in Iran's automotive supply chain, cluster producers faced drastically reduced production due to diminished demand from other SC segments. In this regard, they decided to repurpose their manufacturing and technology facilities to produce emerging products during the pandemic, such as masks and face shields. This operational shift proved that repurposability, as an adaptation strategy, can adjust SC infrastructure to be viable and survive during the pandemic by responding to urgent situational needs. The proposed Repurposable structure can be generalized to the supply chain network design problem and offers a valuable reference for industries where adaptability is the only way to respond to a crisis.

Since repurposing involves costly reconfigurations and is associated with significant uncertainties, it is essential to consider key factors such as the intensity of the super-disruption, the probability of its occurrence, and demand uncertainty when designing an RSC network. The proposed concept is inherently stochastic, involving questions such as: which emerging supply chains can become active during a super-disruption, how demand for each may shift under varying levels of disruption intensity, and what sequences of events are possible across the planning horizon. Given the nature of the problem, a Two-Stage Stochastic Programming (TSSP) approach can capture different types of problem decisions, including repurposing decisions in the first stage and tactical and operational decisions in the second stage. Since the problem is NP-hard in the large, it is necessary to employ an appropriate solution methodology, such as Benders' Decomposition. The impractical aspect of modeling a traditional stochastic programming formulation for the RSC problem using a scenario tree is that the number of possible scenarios grows too large.

Complex scenario trees in stochastic models can be simplified using established methods in the literature, such as Sample Average Approximation (SAA). While these methods have demonstrated their effectiveness, scenario reduction in the context of an RSC can be further enhanced through predictive methods. Estimating scenario probability can be advanced by enabling artificial intelligence (AI), in which the development of digital infrastructure improves data accessibility and enhances the effectiveness of probability estimation. Therefore, these advancements also improve the scenario reduction process. Leveraging AI in this way enables the RSC to be configured more precisely and practically; by harnessing real-time data during super-disruptions, AI facilitates a dynamic and responsive digital RSC configuration.

The present study designs a supply chain network to support operational shifting during a long-term crisis, utilizing the proposed repurposability concept. The key contributions of this research can be summarized as follows:

- A novel mixed-integer linear programming (MILP) model is developed to design an RSC that is capable of shifting its operations as an adaptation strategy during a long-term crisis. In this regard, the proposed RSC is involved in emerging product networks that have emerged of necessity during the crisis, when the primary supply chain has been severely disrupted and cannot function properly. The proposed optimization model determines the repurposing decision for each manufacturer plant individually based on the required repurposing time, variable costs, and capacity.
- The study utilizes a two-stage stochastic programming formulation where the repurposing decision serves as a here-and-now decision. It provides a specific solution and generates greater value than the expected value across possible scenarios. In this stochastic RSC model, the impact level of the long-term crisis in each period plays the most important role in estimating the probabilities and scenarios. In this regard, the study introduces a novel predictive algorithm for the stochastic recourse problem, which intelligently prunes the scenario tree and estimates scenario probabilities using classification-based machine learning. This capability integrates realistic and real-time analysis to enable emerging supply chains during a long-term crisis.
- The stochastic nature and recourse structure of the proposed model indicate that, as the problem size increases, the problem becomes NP-hard and computationally complex. In this regard, the present study develops and applies a solution methodology, Generalized Benders Decomposition (GBD), to effectively and tractably solve the proposed stochastic recourse problem.
- Although investigating responsiveness in an RSC model can be enabled for the main product SC, it is not realistic and sufficient when the SC is involved with emerging product networks. In this regard, the study introduces a multi-objective formulation to identify trade-offs between economic and responsiveness aspects.

The structure of the paper is outlined as follows. Section 2 presents a review of the relevant literature. Section 3 configures the problem statement and mathematical model. Section 4 introduces the proposed learning-based stochastic programming, and Section 5 elaborates on the GBD solution methodology. Section 6 describes the information on the applied case study, and Section 7 discusses the model's computational results. Finally, Section 8 concludes the paper and outlines directions for future research.

2. Literature review

The literature on resilient supply chains can be divided into two distinct perspectives, i.e., a stability and adaptation views (Ivanov, 2024b). The first is the preparedness and recovery capabilities, which focus on supply chains that experience a disruption but begin the recovery process once stability is restored. From this point of view, deviations from planned performance are tried to be minimized, and the system tends toward its pre-disruption state. The second perspective is the adaptation, which addresses supply chains affected by prolonged super-disruptions. In this section, maintaining the system's performance and ensuring long-term viability in a dynamic environment is the primary focus of activities. In other words, the adoption plays an essential role in what is to be done.

2.1. Current trends in preparedness and recovery capabilities

The static view focuses on maintaining a normal state and leveraging redundancies, such as surplus capacity, production, and inventory, to absorb fluctuations. This part of the literature review involves preparedness and the effectiveness of recovery measures in the static view.

Preparedness is the initial one, and most studies have identified widely acknowledged resilience capabilities, such as structural backup diversity (including backup or alternative suppliers), risk-prevention inventory, and flexible capacity. Li et al. (2021) proposed an efficient strategy to address the ripple effect in a supply chain when a disruption affects the order-making process. In this regard, they utilized an integrated strategy that included safety stock, contingent sourcing, and compensating for backorders. Lücker et al. (2021) investigated the optimal allocation of risk mitigation inventory and reserve capacity in multi-stage serial supply chains. Dolgui et al. (2020) amalgamated essential supply chain concepts to augment supply chain flexibility in a disruptive context by introducing dynamic *meta*-structures and autonomous services. Hosseini et al. (2020) simultaneously focused on the supply chain's vulnerability and recoverability by proposing a Bayesian network that led to a new measure. They also proposed managerial strategies aligned with an open-system perspective on resilience. Zheng et al. (2023) examined the impacts of risk prediction in a collective supply chain using a federated learning approach. Their proposed approach investigates order delay before and after the pandemic, where data privacy and accuracy improvement play a critical role.

Some studies have focused on effectiveness and recovery measures, with most of them considering resilience as a measurable attribute of the SC. In this regard, Li et al. (2020) considered the influence of network characteristics on supply chain resilience. They identified the main structural factors and simplified metrics to predict resilience consequences. Torabi et al. (2015) enabled proactive strategies to address the ripple-effect risks in a resilient supply chain by considering supplier selection and order allocation. Also, they proposed a bi-objective two-stage stochastic-possibilistic model to address operational risks. Gholami-Zanjani et al. (2021) included a triad of strategies, readiness, flexibility, and responsiveness, within the food business to alleviate the impacts of disruption from a risk-averse standpoint.

Researchers have examined recovery strategies by evaluating supply chain responses in disrupted environments compared to stable, disruption-free scenarios. Paul and Chowdhury (2021) investigated demand surges and supply shortages due to the pandemic by proposing a model that modifies production plans. Preparedness and adaptive decisions, such as pre-positioning, are made to ensure supply chain resilience during a pandemic in Rozhkov et al. (2022)'s study. Ivanov (2020) examined the impact of facility scheduling and preparation on reducing pandemic risks associated with long-term disruptions and their ripple effects across a supply chain. Proselkov et al. (2024) proposed an agent-based model for firms that are more exposed to risk due to having higher bargaining power and deep-tired suppliers, implying a more critical role in propagating risks. Ivanov (2024a) analyzed how an exit strategy can be effective in supply chains post-COVID-19, where it aids decision-makers in identifying a recovery plan and a disruption tail by proposing a discrete-event simulation. Recently, Yılmaz et al. (2025a) presented a study to consider the potential of an earthquake in Istanbul, which causes demand uncertainty in the humanitarian supply chain network. They proposed a mixed framework, contributing the value stream mapping technique for determining states and a two-stage stochastic programming model for distribution strategies.

The deployment of emergency resilience capabilities is a widely recognized approach in SC resilience literature. Although these capabilities can play an essential role in the resiliency of the supply chains, they may not be operational in all situations. In this regard, Fraccascia et al. (2017) developed a new approach to measuring resilience in industries configured by symbiosis networks by integrating different disciplines' perspectives. Their primary focus was on using waste and having a variety of systems. Aldrighetti et al. (2023) introduced a resilient framework to optimize a trade-off between resiliency and efficiency in multi-echelon SCs, even in non-disrupted scenarios. Alikhani et al. (2021) employed a variety of proactive and reactive resilience strategies to design a resilience network, including inventory prepositioning, reserved capacity, facility fortification, multiple set covering, inventory sharing, and direct-to-store delivery.

As mentioned, all reviewed studies in this subsection implement a recovery plan and employ preparedness measures to mitigate ripple effects. While this approach is practical within an anticipated timeframe, it becomes less capable during prolonged crises or super-disruptions.

2.2. Current trends in the adaptation

The previous subsection examined resilient SC management from a static perspective, focusing on preparedness and recovery measures for responding to disruptions. However, adoption remains a missing link in resilient SC when a return to normal is no longer possible. This concept was introduced by a few previous studies in recent years, in which Ivanov and Dolgui (2020) developed a new insight to assess the viability of super-disruptions at long-term scales by proposing an adoption perspective. Ivanov and Dolgui (2019) relied on designing low-certainty SC networks, which generally focus on decreasing network uncertainty and predictability needs in a resilience framework. In the viability setting, Ivanov and Dolgui (2020) improved SC viability by proposing an intertwined supply network that employs a dynamic game-theoretic approach inspired by biological systems. Furthermore, Ivanov (2022) proposed a new concept, AURA, which emphasizes reducing dependence on predictions and the earned values from assets enabled in a resilience SC.

To guarantee the life of supply chains under extreme disruptions and prolonged crises, viability has emerged as a behavior-driven characteristic of the system (Ivanov and Keskin, 2023) that has garnered considerable interest from scholars in recent years. In this context, Sawik (2023) introduced a stochastic model to preserve SC viability amid ripple effects by optimizing production trajectories to balance cost and service. In a subsequent work, Sawik and Sawik (2024) utilized CVaR-based stochastic optimization to preserve risk-averse supply chain viability amid disturbances, shown through a smartphone manufacturing case study. Münch and Hartmann (2023) examined supply chain resilience during the Covid-19 epidemic using several case studies across three interrelated industries, emphasizing essential ideas such as dynamic and relational reactions to delineate adaptive behavior. The medical supply chain was the most prominent section, which researchers investigated during the COVID-19 epidemic. In this regard, two research studies, including Yılmaz et al. (2025b) and Özçelik et al. (2025), considered the medical products allocation problem under uncertainty during the COVID-19 epidemic to assess the viability level.

Furthermore, Alizadeh et al. (2023) designed a viable pandemic healthcare network to maximize recovery, minimize costs, and reduce death rates by developing a multi-stage stochastic mathematical model. Brusset et al. (2023) examined supplier selection and

Table 1
A summary of the literature review.

References	Preparedness	Recovery	Adoption	Modeling Approach	Empowered by Digital Twin	Decision-Making	Solution Methodology
Li et al. (2021)	✓	–	–	Optimal Control	–	–	–
Lücker et al. (2021)	✓	–	–	Mathematical Model	–	Risk-Averse	Derivation
Hosseini et al. (2020)	✓	✓	–	Bayesian Network	–	Risk-Averse	Likelihood Analysis
Zheng et al. (2023)	✓	–	–	Learning Approach	✓	Risk-Averse	–
Torabi et al. (2015)	–	✓	–	MILP Mathematical Model	–	two-stage stochastic-possibilistic	*meta*-heuristic
Paul and Chowdhury (2021)	–	✓	–	Mathematical Model	–	–	Generalized Reduced Gradient
Rozhkov et al. (2022)	✓	✓	–	Simulation Model	–	Scenario-based model	–
Ivanov (2020)	✓	✓	–	Simulation Model	–	Scenario-based model	–
Ivanov (2024a)	–	✓	–	Simulation Model	–	Scenario-based model	–
Aldrighetti et al. (2023)	✓	✓	–	MINLP Mathematical Model	–	two-stage stochastic	Benders and Local Search
Ivanov and Dolgui (2020)	–	–	✓	Game-theoretical Model	–	Risk-Averse	–
Sawik and Sawik (2024)	–	–	✓	MIQP Mathematical Model	–	CvaR stochastic	–
Alizadeh et al. (2023)	–	–	✓	MILP Mathematical Model	–	Two-stage stochastic	–
Brusset et al. (2023)	–	–	✓	Optimal Control	–	Scenario-based model	–
Sardesai and Klingebiel (2023)	–	–	✓	Process Capability Index	–	–	–
Zhu et al. (2024)	–	–	✓	Agent-based model	✓	–	–
Kumar et al. (2023)	–	–	✓	Structural Equation Modeling	✓	–	–
Dolgui et al. (2024)	–	–	✓	MILP Mathematical Model	–	Scenario simulation	–
Present Study	✓	–	Repurposable SC	MILP Mathematical Model	Learning-based TSSP	TSSP	GBD

order planning within a supply chain network experiencing disruption by including network reconfigurations and pandemic dynamics, employing an optimal control model.

Technological advancements and the advent of digital twin methodologies have significantly enhanced the viability of SCs during prolonged crises, rendering them leaner and more resilient. In this regard, Sardesai and Klingebiel (2023) suggested a quantitative way using a digital twin approach that uses a mixed resilience index to look at the amount of SC resilience. They evaluated potential reconfigured SCs using the proposed index to select the most effective one based on the process capability index for long-term recovery planning. Zhu et al. (2024) utilized behavioral principles, such as win–win interaction, cooperation establishment, and cooperation prioritizing, to attain viability in a heterogeneous supply system through the development of an agent-based model. Kumar et al. (2023) investigated the roles of digitalization and viability in enriching SC capabilities to address disruption risks, where they control ripple effects by building resilience to gain a competitive advantage, drawing on the resource-based view.

Dolgui et al. (2024) introduced the concept of repurposable SC in a different framework, the network-of-networks, and developed a manufacturing adaptation approach using scenario simulation. Although the study presents the concept of repurposing, it ignores several critical aspects, including:

- As it is clear, repurposing for manufacturer facilities generally imposes substantial costs on the system and business. In this regard, an optimization procedure must be conducted to determine repurposing decisions for each assembly plant based on the required repurposing time, variable costs, and capacity.
- If the disruptions have known characteristics and recovery scenarios are calculable, deterministic models can be practical, but they are not always applicable. It is necessary to propose an algorithm that estimates the impact-level probabilities of long-term crises in each period.
- Although scenario simulation has played an effective role in estimating practical scenarios for many years, it is less useful for long-term, large-scale crises. In contrast, a stochastic formulation provides a specific solution and generates greater value than the expected value across possible scenarios in recourse models.
- Since repurposing in SCs is a binary decision, and we likely face an MILP mathematical model, it is an NP-hard problem in large-scale networks, and we need a solution methodology when commercial solvers are often impractical.
- Investigating responsiveness for repurposable manufacturing alone is unrealistic for emerging products, so proposing a multi-objective formulation to identify trade-offs between economic and responsiveness aspects is critical.

Table 1 presents a summary of the most prominent and relevant studies reviewed in this section.

By reviewing Table 1, we identified critical gaps in the literature that this study addresses most significantly. In the first step, the study proposes a repurposable concept in SC that enhances SC's adaptability to long-term crises, thereby maintaining survivability. This proposed idea changes the direction of previous studies, which have often sought to achieve viability through redundancy-resilience measures. In the second step, by integrating the proposed network with a novel predictive stochastic formulation, the viability of RSC is most effectively enhanced based on a digital and AI manner. At the end, since commercial solvers are incapable of solving the RCS problem at scale, the study addresses it by developing and implementing a solution method based on the Generalized Benders Decomposition.

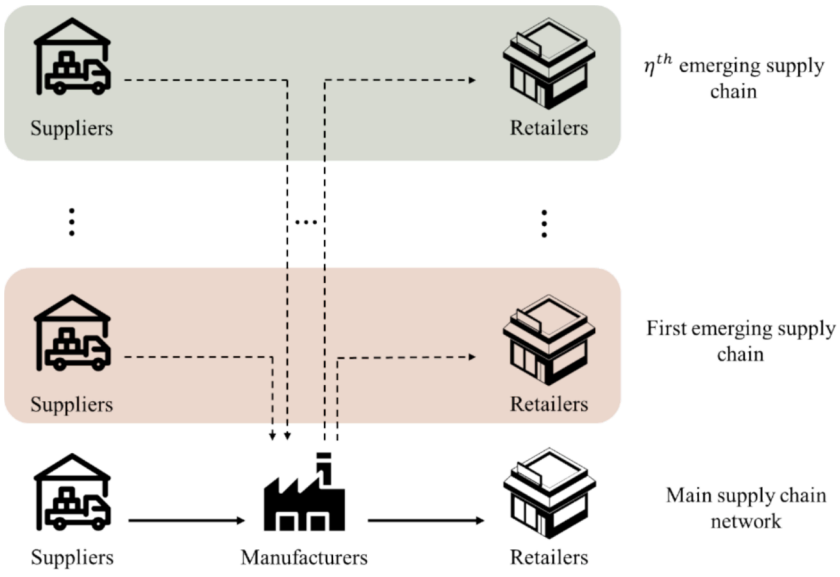


Fig. 1. Schematic representation of the supply chain adaptation with emergent networks under long-term crisis.

3. Problem statement

In recent years, long-term crises have disrupted SCs worldwide, including the COVID-19 pandemic, the global semiconductor shortage, and the Russia-Ukraine conflict (Gilani et al., 2025a). In such situations, the concept of a viable supply chain becomes crucial in SCM. A viable supply chain management focuses on ensuring continued SC operations and the SC's survivability when a crisis threatens the SC's existence. One of the most effective ways to maintain a lean SC and profitability during a super-disruption is repurposing. This concept implies adapting the SC during a long-term crisis to address emerging needs. For instance, Ford repurposed existing equipment and technology across its assembly plants and supplier networks to leverage emerging product networks during the COVID pandemic, such as face shields and ventilators.

Fig. 1 shows the main SC network, including suppliers in the first layer, manufacturers in the second layer, and retailers in the third layer. Since this network has been exposed to a long-term crisis, some product networks have emerged from necessity, allowing the main SC repurpose itself to remain viable and survive.

When a long-term crisis has minimal impact, the condition of SC is normal, so the primary SC proceeds with its operations as usual, meaning raw materials are delivered to manufacturers and finished products are sent to retailers. If impacts grow to a medium or high level, usual operations will break down. Thus, some critical parts of SC may not work as usual or correctly, for example, due to a lack of raw materials or an imbalance in final product demand. In these situations, an RSC becomes essential for survival. In these crises, RSC plays a vital role in survivability, identifying and shifting to product networks that emerge from necessity during the situation. This adaptive behavior ensures the SC maintains viability and survivability during long-term crises by keeping its operations continued.

A response to this challenge can be designing a repurposable SC that adapts to the severity of a long-term crisis, while leveraging new, emerging product networks that they formed of necessity by integrating itself into them. The present study aims to develop a repurposable network that is capable of shifting its operations across different crisis stages. Additionally, the model must optimize facility capacities, material flows, production schedules, and inventory levels to satisfy demand in both the original and emergent SCs. Fig. 2 illustrates an RSC that integrates into two emerging product markets by participating in their respective supply networks.

This problem is modeled based on several key assumptions, including:

- Each manufacturer can be designated for repurposing to support only a single emerging SC. Since efforts for equipment setting are time-consuming and costly, it is only enabled for an emerging supply network.
- Repurposing decision is considered as a binary decision in each manufacturer for each emerging network. This shows it is a strategic decision that SC managers have to make only once in each time horizon, according to the costs and benefits.
- Lead times for raw material orders are considered exclusively in the emerging SCs and are not applied to the main SC. It is clear that the production procedure for the main SC is executing, but running new lines for the emerging SC needs to order new raw material.
- The repurposing process at manufacturers requires a defined time duration to be completed before production can begin. Changing settings and relocation facilities need a specific time before running an emerging SC.
- Inventory storage is allowed solely at manufacturer locations. Since there is a main focus on manufacturers in the proposed idea, the inventory level is controlled in this facility.
- While meeting demand in the main SC is mandatory, fulfilling demand in the emerging SCs is considered optional. In other words, there is no necessity that repuposable network satisfy all demand in emerging networks.

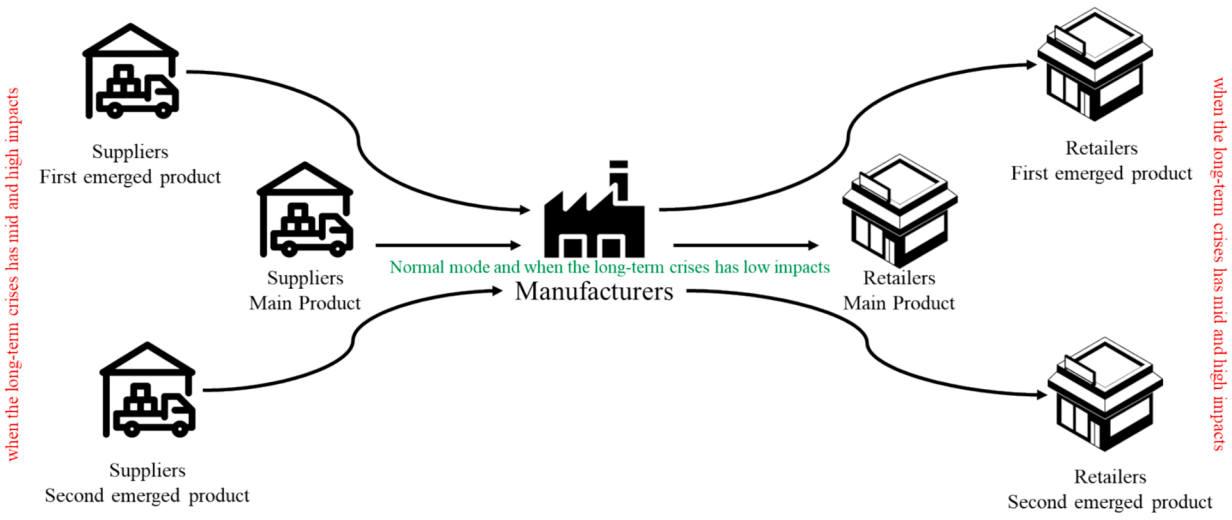


Fig. 2. Visualization of an RSC connected to two emerging product markets via dedicated supply networks.

3.1. Repurposable supply chain mathematical modeling

This section presents the RSC model. First, the model's notations are listed, including sets, indices, parameters, and variables. Then, the objective functions and operational constraints are formulated.

Sets	
I	Set of suppliers
J	Set of manufacturers
R	Set of raw materials
P	Set of final products
K	Set of retailers
H	Set of supply chains emerging during long-term crises
S	Set of scenarios
T	Set of time horizon
Indices	
i_0	Index of suppliers in the main supply chain ($i_0 \in I$)
i_h	Index of suppliers in the hth emerging supply chain ($i_h \in I$)
j	Index of manufacturers ($j \in J$)
r_0	Index of raw materials in the main supply chain ($r_0 \in R$)
r_h	Index of raw materials in the hth emerging supply chain ($r_h \in R$)
p_0	Index of final products in the main supply chain ($p_0 \in P$)
p_h	Index of final products in the hth emerging supply chain ($p_h \in P$)
k_0	Index of retailers in the main supply chain ($k_0 \in K$)
k_h	Index of retailers in the hth emerging supply chain ($k_h \in K$)
h	Index of supply chains emerging during long-term crises ($h \in H$)
s	Index of scenarios ($s \in S$)
t	Index of periods ($t \in T$)
Parameters	
u_{jh}	Repurposing costs for manufacturer j in emerged supply chain h
q_r	Unit transportation cost of raw material r per unit distance
q_p	Unit transportation cost of final product p per unit distance
v_p	Unit variable cost per unit of capacity for product p at manufacturers
υ_p	Unit variable cost per unit of expansion capacity for product p at manufacturers
m_r	Unit purchasing cost of raw material r from a supplier
l_p	Unit production cost of product p at manufacturers
$\mathscr{K}_p$	Unit inventory cost of product p at a manufacturer
dr_{ij}	Distance from node i to node j
dp_{jk}	Distance from node j to node k
$\mathscr{D}_{kp}^{st}$	Demand of product p in retailer k under scenario s in period t
$\overline{G}_{ir}$	Maximum available quantity of raw material r at supplier i
μ_i^{st}	Binary parameter indicating whether supplier i is available in scenario s during period t
$\overline{C}_p$	Maximum available designed capacity for product p in each manufacturer
α_{rp}	Coefficient representing the amount of raw material r required to produce one unit of product p
M	A big number
Ld_i	Lead time for procuring raw material from supplier i in the emerging supply chain
η_j	Repurposing time at the manufacturer j
Pos_s	Probability of scenario s
Binary variables	
β_{jh}	If the manufacturer j is selected for repurposing of the emerged supply chain h 1, otherwise 0.
Continuous variables	
x_{ijr}^{st}	Transported raw material r from supplier I to manufacturer j under scenario s in period t for the main supply chain
π_{ijrh}^{st}	Transported raw material r from supplier I to manufacturer j under scenario s in period t for the emerged supply chain h
y_{jkp}^{st}	Amount of transported product p from manufacturer j to retailer k under scenario s in period t for the main supply chain
σ_{jkph}^{st}	Amount of transported product p from manufacturer j to retailer k under scenario s in period t for the emerged supply chain h
c_{jp}^{st}	Designed capacity of product p at manufacturer j under scenario s in period t for main supply chain products
ρ_{jph}^{st}	Designed capacity of product p at manufacturer j under scenario s in period t for emerged supply chain products h
λ_{jph}^{st}	Expansion capacity of product p at manufacturer j under scenario s in period t for the emerged supply chain h
N_{jp}^{st}	Amount of product p produced at manufacturer j under scenario s in period t for the main supply chain
γ_{jph}^{st}	Amount of product p produced at manufacturer j under scenario s in period t for the emerged supply chain products h
χ_{jp}^{st}	Inventory of product p at manufacturer j under scenario s in period t for the main supply chain
w_{jph}^{st}	Inventory of product p produced at manufacturer j under scenario s in period t for the emerged supply chain products h

- Objective functions

Eq. (1) formulates the first objective function of the proposed RSC model. This objective function involves the total operational network costs. The equation includes terms that represent parts of the expenses. The first and second terms imply transportation costs for raw materials and finished manufactured products, respectively. The third term represents variable costs associated with capacity

utilization for each manufacturer. The fourth term represents the costs of capacity expansion resulting from increased production during long-term crises. The fifth term covers raw material procurement costs, while the sixth represents production costs for both main and emerging products in their respective SCs. Finally, the objective includes inventory holding costs at manufacturers for both main and emerging products.

$$\begin{aligned}\min Z_1 = & \underbrace{\sum_j \sum_h u_{jh}\beta_{jh}}_{\text{Repurposing costs}} + \underbrace{\sum_j \sum_s \sum_t \left[\left(\sum_{r\in r_0} \sum_{i\in i_0} dr_{ij}.q_r.x_{ijr}^{st} \right) + \left(\sum_{r\in r_h} \sum_{i\in i_h} dr_{ij}.q_r.\pi_{ijrh}^{st} \right) \right]}_{\text{costs of raw material transportation}} \\ & + \underbrace{\sum_j \sum_s \sum_t \left[\left(\sum_{k\in k_0} \sum_{p\in p_0} dp_{jk}.q_p.y_{jkp}^{st} \right) + \left(\sum_{k\in k_h} \sum_{p\in p_h} \sum_h dp_{jk}.q_p.\sigma_{jkph}^{st} \right) \right]}_{\text{costs of product transportation}} \\ & + \underbrace{\sum_j \sum_s \sum_t \left[\left(\sum_{p\in p_0} c_{jp}^{st}.v_p \right) + \left(\sum_{p\in p_h} \sum_h \rho_{jph}^{st} v_p \right) \right]}_{\text{variable costs}} + \underbrace{\sum_j \sum_s \sum_t \left(\sum_{p\in p_h} \sum_h \lambda_{jph}^{st} v_p \right)}_{\text{capacity expansion costs}} \\ & + \underbrace{\sum_j \sum_s \sum_t \left[\left(\sum_{i\in i_0} \sum_{r\in r_0} x_{ijr}^{st} m_r \right) + \left(\sum_{i\in i_h} \sum_{r\in r_h} \sum_h \pi_{ijrh}^{st} m_r \right) \right]}_{\text{Purchasing costs}} \\ & + \underbrace{\sum_j \sum_s \sum_t \left[\left(\sum_{p\in p_0} N_{jp}^{st}.l_p \right) + \left(\sum_{p\in p_h} \sum_h \gamma_{jph}^{st} l_p \right) \right]}_{\text{Production costs}} \\ & + \underbrace{\sum_j \sum_s \sum_t \left[\left(\sum_{p\in p_0} \chi_{jp}^{st}.\mathscr{H}_p \right) + \left(\sum_{p\in p_h} \sum_h w_{jph}^{st} \mathscr{H}_p \right) \right]}_{\text{Inventory costs}}\end{aligned} \tag{1}$$

$$\min Z_2 = \sum_{k\in k_h} \sum_{p\in p_h} \sum_h \sum_s Pos_s \left(\frac{\sum_j \sum_t \sigma_{jkph}^{st}}{\sum_t \mathscr{D}_{kp}^{st}} \right) \tag{2}$$

Eq. (2) defines the second objective function of the RSC model, which maximizes the proportion of fulfilled demand for emerging products over the entire time horizon, enhancing the performance of emerging SCs during long-term crises.

- Constraints on supplying raw material

Eqs (3) and (4) define the supply of raw materials from suppliers to the main SC and the emerging SCs, respectively. In these equations, $\overline{G}_{ir}$ represents the maximum available quantity of raw material r that supplier i can provide, while μ_i^{st} is a binary parameter that is introduced as follows:

$$\mu_i^{st} = \begin{cases} 0, & \text{if supplier } i \text{ under scenario } s \text{ in period } t \text{ is not available} \quad \forall i \in I,\ s \in S \\ 1 & \text{, otherwise} \end{cases}$$

Accordingly, the flow of raw materials is constrained by both the supplier's availability and their maximum supply capacity. Ld_i represents the lead time required to supply raw materials from suppliers for the production of emerging products.

$$\sum_j x_{ijr}^{st} \le \overline{G}_{ir} \mu_i^{st} \quad \forall i \in i_0,\ r \in r_0, s, t \tag{3}$$

$$\sum_j \pi_{ijrh}^{s,t+Ld_i} \le \overline{G}_{ir} \mu_i^{st} \quad \forall i \in i_h, r \in r_h, h, s, t \tag{4}$$

- Constraints on manufacturer capacity

Manufacturer capacities are limited by a maximum allowable value in each period, denoted by $\overline{C}_p$. This capacity applies to both main and emerging products across the main and emerging SCs. However, for the emerged SCs, production is only permitted if the manufacturer has been selected for repurposing. These constraints are formulated in Eqs (5) and (6).

$$c_{jp}^{st} \leq \overline{C}_p \quad \forall j,p \in p_0,s,t \tag{5}$$

$$\rho_{jph}^{st} \leq \overline{C}_p \beta_{jh} \quad \forall j,p \in p_h,h,s,t \tag{6}$$

- Constraints on production limit

Manufacturers produce both main and emerging products within their designed capacity limits. In Equations (7) and (8), N_{jp}^{st} and γ_{jph}^{st} represent the quantities of main and emerging product production at each manufacturer, respectively, for the main and emerging SCs.

$$N_{jp}^{st} \leq c_{jp}^{st} \quad \forall j,p \in p_0,s,t \tag{7}$$

$$\gamma_{jph}^{st} \leq \rho_{jph}^{st} \quad \forall j,p \in p_h,h,s,t \tag{8}$$

- Balance constraints

As manufacturers can hold inventory, balance constraints are formulated for both the main and emerging SCs. IN_{jp}^{st} and w_{jph}^{st} denote the inventory levels of main and emerging products, respectively. Equations (9) and (10) are based on the inventory balance principle: the inventory level of the current period equals the previous period's inventory plus production minus the quantity delivered.

$$\chi_{jp}^{s,t-1} + N_{jp}^{st} - \sum_{k \in k_0} y_{jkp}^{st} = \chi_{jp}^{st} \quad \forall j,p \in p_0,s,t \tag{9}$$

$$w_{jph}^{s,t-1} + \gamma_{jph}^{st} - \sum_{k \in k_h} \sigma_{jkph}^{st} = w_{jph}^{st} \quad \forall j,p \in p_h,h,s,t \tag{10}$$

- Assembly constraints

Eqs (11) and (12) represent the assembly constraints, where α_{rp} denotes the amount of raw material r required to produce one unit of product p. These constraints are applied to both main and emerging products. However, it is essential to note that the production of emerging products requires repurposing time, represented by η_j.

$$\frac{\sum_j x_{ijr}^{st}}{\alpha_{rp}} \geq N_{jp}^{st} \quad \forall j,p \in p_0,r \in r_0,s,t \tag{11}$$

$$\frac{\sum_j \pi_{ijrh}^{st}}{\alpha_{rp}} \geq \gamma_{jph}^{s,t+\eta_j} \quad \forall j,p \in p_h,r \in r_h,h,s,t \tag{12}$$

- Capacity expansion constraints

Eq. (13) determines the capacity expansions of manufacturers for producing emerging products, where λ_{jph}^{st} represents the capacity expansion of manufacturer j for product p in the emerging supply chain h. It is important to note that such expansions are not considered for main products.

$$\rho_{jph}^{st} = \rho_{jph}^{s,t-1} + \lambda_{jph}^{st} \quad \forall j,p \in p_h,h,s,t \tag{13}$$

- Constraints on meeting demand

Meeting demand is mandatory for main products but optional for emerging products. Accordingly, Eq. (14) ensures demand fulfillment in the main SC, while Eq. (15) applies to the emerged SCs. Here, $\mathscr{D}_{kp}^{st}$ represents the demand for product p at retailer k.

$$\sum_j y_{jkp}^{st} \geq \mathscr{D}_{kp}^{st} \quad \forall k \in k_0,p \in p_0,s,t \tag{14}$$

$$\sum_{j} \sigma^{st}_{jkph} \leq \mathscr{D}^{st}_{kp} \quad \forall k \in k_h, p \in p_h, h, s, t \tag{15}$$

- Repurposing constraints

Eq. (16) ensures that raw materials for emerged products are delivered to a manufacturer only if it has been selected for repurposing, where M is a sufficiently large constant. Eq. (17) restricts each manufacturer to repurpose for only one emerged SC.

$$\pi^{st}_{ijrh} \leq \mathrm{M}\beta_{jh} \quad \forall i \in i_h, j, r \in r_h, h, s, t \tag{16}$$

$$\sum_{h} \beta_{jh} \leq 1 \quad \forall j \tag{17}$$

3.2. Dealing with Multi-Objective optimization trade-offs

As formulated earlier, the proposed model has two objective functions, so we face a multi-objective mathematical model that generally does not admit a unique solution. In this regard, it is clear to identify the Pareto front. One of the most common methods to find a Pareto solution is the ε-constraint method. This method plays a significant role in solving multi-objective mathematical models, where an objective function serves as the primary objective, and the remaining objective functions appear as constraints limited by a threshold parameter, ε (Gilani et al., 2023, Aghajani et al., 2025). By systematically varying ε, a set of solutions, known as the Pareto front, is produced, and decision makers choose their most preferred solution according to their preferences (Laumanns et al., 2006, Gilani and Sahebi, 2024).

Assume a general multi-objective model as follows:

$$\min_{x \in X} (f_1(x), f_2(x), \cdots, f_k(x))$$

$$\text{subject to } x \in X \subseteq \mathbb{R}^n$$

Where $f_1(x) : \mathbb{R}^n \rightarrow \mathbb{R}$ for $i = 1, 2, \cdots, k$ are the objective functions, and X is the feasible solution space defined by a set of equality and/or inequality constraints.

In the ε-constraint method, a single objective function is chosen as the master objective, and the rest of the $k-1$ objectives are included in the problem as constraints bound by user-supplied threshold parameters ε_i (Alipoor et al., 2024). Without loss of generality, let $f_i(x)$ be taken as the master objective. The ε-constrained problem is redefined as:

$$\min_{x \in X} f_1(x)$$

$$\text{subject to } \; f_i(x) \leq \varepsilon_i \quad i = 1, 2, \cdots, k$$

$$x \in X$$

The choice of ε-values $(\varepsilon_2, \varepsilon_3, \cdots, \varepsilon_k)$ governs the trade-off between objectives. An approximation to the Pareto frontier can be found by solving a series of such problems with systematically varying ε-values.

4. Learning-based Two-Stage stochastic programming

As mentioned in the last section, the suggested model is a mixed-integer linear recourse model that uses binary decision variables, such as repurposing decisions. Some parameters in this model are subject to uncertainty. Accordingly, a stochastic mixed-integer linear programming formulation with random parameters in the constraints is presented as follows:

$$\min \begin{bmatrix} Ax = b \\ cx : \; T(\omega)x \; h(\omega) \\ x \in X \end{bmatrix} \tag{18}$$

The set X defines both the lower and upper bounds of the decision variables, as well as their respective types, as outlined below:

$$X = \left\{ x \in \mathbb{Z}^{\overline{n}} \times \mathbb{R}^{n-\overline{n}} : x^l \leq x \leq x^u \right\}$$

Here, $\bar{n}$ indicates which components of the n-dimensional vector x are integer variables. The constraint $Ax = b$ the deterministic part of the model, while $T(\omega)x\ h(\omega)$ captures the constraints that depend linearly on the random vector $\omega \in \mathbb{R}^r$. Before ω happens, decisions have to be taken, and the only thing that can be known is probability of occurrence. One of the most common models in this context is the recourse-based structure, where q represents the recourse cost, W is the recourse matrix, and the set Y defines simple bounds and variable type restrictions ($y \in \mathbb{Z}^{\bar{p}} \times \mathbb{R}^{p-\bar{p}} : y^l \le y \le y^u$). The canonical form of the problem is as follows:

$$\min_{x\in X}\left\{ \underbrace{cx}_{First-stage} + \mathbb{E}_{\omega}\left[\underbrace{\min_{y\in Y}\{qy : Wy\ h(\omega) - T(\omega)x\}}_{Second-stage} \right] : Ax = b \right\} \tag{19}$$

In this formulation, the first-stage decisions incorporate the decisions that must be made before the realization of uncertainty, named here-and-now. The second-stage decisions are short-term decisions that depend on the values of the random vector's realization, named wait-and-see (Siddig and Song, 2025).

The expected value expression in the second stage of the above formulation, which implies the random vector, is a critical part that needs to be considered. By considering the random vector within a discrete space, the formulation can be represented using a scenario-based model, where pos_s denotes the probability of scenario $s \in \omega$, as shown below:

$$\min_{x\in X}\left\{ \underbrace{cx}_{First-stage} + \sum_{s\in\omega} pos_s \left[\underbrace{\min_{y\in Y}\{qy : Wy\ h(s) - T(s)x\}}_{Second-stage} \right] : Ax = b \right\} \tag{20}$$

According to the above formulations, the final two-stage RSC model can be reformulated as follows:

$$\begin{aligned}
\min Z_1 = {} & \underbrace{\sum_j \sum_h u_{jh}\beta_{jh}}_{Repurposing\ costs} + \underbrace{\sum_j \sum_s \sum_t Pos_s \left[\left(\sum_{r\in r_0} \sum_{i\in i_0} dr_{ij}.q_r.x^{st}_{ijr} \right) + \left(\sum_{r\in r_h} \sum_{i\in i_h} dr_{ij}.q_r.\pi^{st}_{ijrh} \right) \right]}_{costs\ of\ raw\ material\ transportation} \\
& + \underbrace{\sum_j \sum_s \sum_t Pos_s \left[\left(\sum_{k\in k_0} \sum_{p\in p_0} dp_{jk}.q_p.y^{st}_{jkp} \right) + \left(\sum_{k\in k_h} \sum_{p\in p_h} \sum_h dp_{jk}.q_p.\sigma^{st}_{jkph} \right) \right]}_{costs\ of\ product\ transportation} \\
& + \underbrace{\sum_j \sum_s \sum_t Pos_s \left[\left(\sum_{p\in p_0} c^{st}_{jp}.v_p \right) + \left(\sum_{p\in p_h} \sum_h \rho^{st}_{jph} v_p \right) \right]}_{variable\ costs} + \underbrace{\sum_j \sum_s \sum_t Pos_s \left(\sum_{p\in p_h} \sum_h \lambda^{st}_{jph} v_p \right)}_{capacity\ expansion\ costs} \\
& + \underbrace{\sum_j \sum_s \sum_t Pos_s \left[\left(\sum_{i\in i_0} \sum_{r\in r_0} x^{st}_{ijr} m_r \right) + \left(\sum_{i\in i_h} \sum_{r\in r_h} \sum_h \pi^{st}_{ijrh} m_r \right) \right]}_{Purchasing\ costs} \\
& + \underbrace{\sum_j \sum_s \sum_t Pos_s \left[\left(\sum_{p\in p_0} N^{st}_{jp}.l_p \right) + \left(\sum_{p\in p_h} \sum_h \gamma^{st}_{jph} l_p \right) \right]}_{Production\ costs} \\
& + \underbrace{\sum_j \sum_s \sum_t Pos_s \left[\left(\sum_{p\in p_0} \chi^{st}_{jp}.\mathscr{H}_p \right) + \left(\sum_{p\in p_h} \sum_h w^{st}_{jph} \mathscr{H}_p \right) \right]}_{Inventory\ costs}
\end{aligned} \tag{16-A}$$

$$\min Z_2 = \sum_{k\in k_h} \sum_{p\in p_h} \sum_h \sum_s Pos_s \left(\frac{\sum_j \sum_t \sigma^{st}_{jkph}}{\sum_t \mathscr{D}^{st}_{kp}} \right) \tag{17-A}$$

$Constraints(3) - (17)$

$$\beta_{jh} \in \{0,1\}, x^{st}_{ijr}, \pi^{st}_{ijrh}, y^{st}_{jkp}, \sigma^{st}_{jkph}, c^{st}_{jp}, \rho^{st}_{jph}, \lambda^{st}_{jph}, N^{st}_{jp}, \gamma^{st}_{jph}, \chi^{st}_{jp}, w^{st}_{jph} \ge 0, \quad \forall i,j,r,p,k,s,t$$

The final formulation of the RSC model incorporating the ε-constraint method is presented in the Appendix.

However, estimating its probability distribution is often challenging in real-world applications, since the realized data typically exhibit a mixed structure and rarely conform to a single, well-defined distribution. This challenge manifests in two main areas:

- Estimating the probability of disruption occurrence
- Determining scenario occurrence probabilities within the scenario tree

In addition to the above challenge, forming a scenario tree in real-world applications involves numerous estimated scenarios due to combinations of disruption intensities across different periods. To address scenario complexity, methods such as SAA can prune the scenario tree and reduce the number of possible scenarios.

4.1. Role of real-time data in two-stage stochastic programming

As mentioned earlier, when historical data do not obey a specific distribution, estimating the likelihood of a disruption event and the probability of an occurred scenario will be challenging. On the other hand, since there are several complexities in the disruption event frequency within a given period, the scenario tree tends to have more branches and be more complex. This complicated scenario tree cannot be enabled in a mathematical model simply because it affects the problem scale and size.

The data era, characterized by the rise of technological gadgets, the ability to store high-dimensional data, and the emergence of artificial intelligence algorithms, has made prediction models more accurate and reliable. In some types of these algorithms, it does not matter whether historical data follow a specific distribution when estimating probabilities, unlike in traditional statistical methods. These advancements aid decision-makers, especially in scenario-based models such as stochastic programming, when historical data on disruptive events do not follow a known distribution.

Machine learning techniques, a big part of artificial intelligence, may help stochastic programming in two primary ways. Stochastic models usually depend on the likelihood of disruptive events; classification algorithms can be used to predict whether a disruptive event will occur (class 1) or not (class 0). These algorithms also give an estimate of the chance of class membership, which helps people make better decisions. As a result, decision-makers may use these tools to predict the likelihood of a disruption or scenario accurately and to reduce the scenario tree.

To utilize these capabilities, it is assumed that a long-term crisis can result in three levels of impact on the system: low, medium, and high. Based on this assumption, a scenario tree over a time horizon of T periods can be constructed, as illustrated in Fig. 3.

As previously noted, when historical data is limited and does not follow a specific distribution, the probability of occurrence for each impact level and, consequently, for each scenario must be estimated. A scenario is defined as a sequence of impact levels over the time horizon. As shown in Fig. 3, the scenario tree contains 3^T possible scenarios, where each period can experience one of three impact levels. For example, a six-period planning horizon with three levels of impact results in 729 possible scenarios. This number significantly increases the model's complexity and places it within the class of NP-hard problems. To address this, a data-driven scenario reduction technique is required to obtain a manageable and practical model.

To solve this problem, Algorithm 1 was designed to determine the probabilities of each disruption-impact level and, by extension, of each scenario. The algorithm also reduces the number of scenarios by eliminating those that are less likely to occur. This makes the

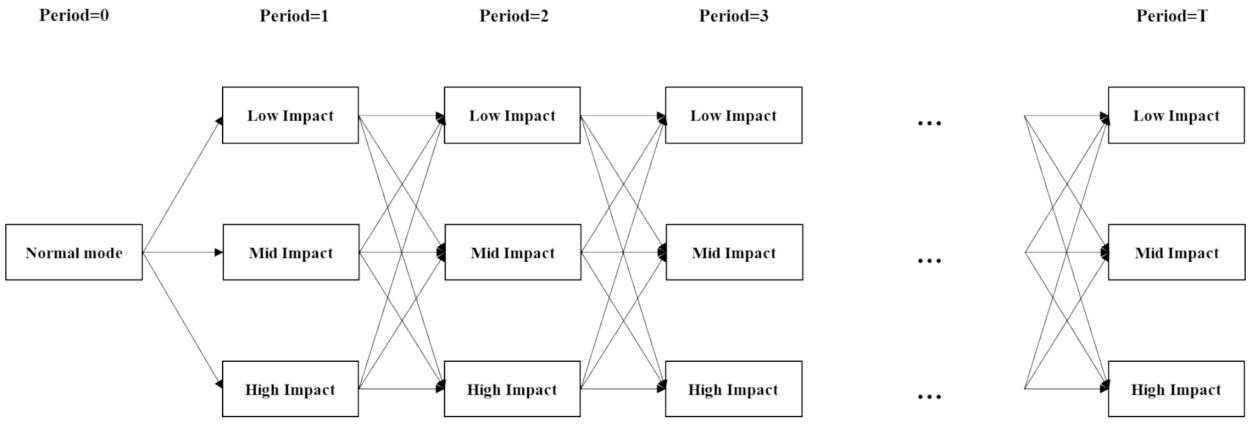


Fig. 3. Scenario tree representation across a time horizon of T periods.

model easier to work with.

Algorithm1. Predicting probability of occurrence and Data-Driven scenario reduction

Input: $dataset^t$,
Output: $\vec{p}_c$
For $\acute{t} \in T$
  For $\acute{c} \in C$
    Split dataset for $t = \acute{t}$
    Train the logistic regression model φ on $dataset^{\acute{t}}$
    Predict $p_{ic}^{\acute{t}}$ (Probability for each datapoint)
    Calculate $\frac{1}{N^{\acute{t}}} \sum_{i=1}^{N^{\acute{t}}} p_{ic}^{\acute{t}}$
    Return $\vec{p}_{c=\acute{c}}^{t=\acute{t}}$
  End
End
For $\acute{c} \in C$
  For $\acute{t} \in T$
    Set $\varepsilon_{\acute{c}}$= threshold
    If $\vec{p}_{c=\acute{c}}^{t=\acute{t}} \geq \varepsilon_{\acute{c}}$ then
      Hold node in scenario tree
    else: Remove node in scenario tree
    End
  End
End

This algorithm takes a dataset as input and outputs the probability of being in a given impact level of a long-term crisis. The dataset is then split into the related time periods according to the mathematical model's time horizon. Next, a multi-class logistic regression model is trained on the data to figure out how likely it is that each data point belongs to each impact level. Finally, for each class, the average probability across all data points is found. This gives us the estimated vector $\vec{p}_c$. This is the first loop of the algorithm that was suggested.

The algorithm's goal in the second loop is to cut down on the number of possible scenarios by using a threshold-based system. There is a set threshold for the probability of each impact level. If the anticipated likelihood of a particular impact level (e.g., medium impact) is below this threshold, the node associated with that level is not included in the scenario tree. The remaining nodes incorporate the smaller set of possible scenarios. If we assume that time periods are independent, we can compute the likelihood of any reduced scenario by multiplying the probabilities of its different effect levels over the time horizon. The pseudo-code for the suggested learning-based stochastic programming method and scenario reduction process in this work is shown below.

Regarding the proposed algorithm, it is mentioned that the algorithm assumes temporal independence across impact levels to compute scenario probabilities. This assumption is enabled in the proposed algorithm according to the characteristics of the suggested long-term crisis. Since the dataset is generally limited and noisy in regarding of a particular long-term crisis, there are some implications indicate that applied independence assumption is reasonable, including: 1) estimating correlations generally accompanies with high level of errors in such long-term crisis (there is a significant uncertainty), 2) entering such unprecise correlations often leads to an unstable model which has intensely fluctuated behavior, 3) according to the unique characters of such long-term crisis, the

independence models generally indicates a performance with more conservatism and transparency which it transforms to a tractable model.

Pseudo-code of the proposed learning-based approach for scenario reduction and probability estimation

```
Begin
Identify and categorize the potential impacts of a long-term crisis
Develop a scenario tree structured according to impact levels across each time period
Extract historical data across various features and time periods
Apply a classification machine learning algorithm
Predict the probability of each data point belonging to each impact level (class)
Estimate the probability of each new data point belonging to each impact level
Prune low-probability nodes in accordance with Algorithm 1
    If Number of remained scenarios is reasonable then
        Construct a new reduced scenario tree based on the remaining scenarios
    else: If Number of remaining scenarios is high then
            Decrease the value of ε and execute Algorithm 1
        If else: Number of remaining scenarios is low then
            Increase the value of ε and execute Algorithm 1
        else: go to the next step
        End
    End
Construct a new reduced scenario tree based on the remaining scenarios
Compute scenario probabilities assuming independence among events
End
```

As mentioned, the logistic regression is the machine learning model that we enabled in the proposed algorithm. Although different machine learning models predict the probability of being in a given impact level according to the accessible dataset, the logistic regression has some characteristics that distinguish it from other models. First, the logistic regression does not need data with a normal distribution, such as Linear Discriminant Analysis. Second, in contrast to some models, it shows less dependence on regularization. Third, it is able to predict the belonging probability of each class, in contrast to the Support Vector Machine. Fourth, it has lower complexity and high speed to train from data related to other models, where we need to make decisions promptly and have a limited dataset. Fifth, it is a transparent model in contrast to Random Forest and some Neural Network models, which are generally black-box. Sixth, it is more practical when there are no nonlinear interactions between the features.

4.2. Logistic regression as a model for class membership probability

A regression model used for classification when the dependent variable has more than two classes is Multinomial logistic regression, also known as softmax regression. It can be used to estimate the probability of a disruptive event occurring at one of the different impact levels. The model outputs a disruption occurrence probability for each possible disruption impact level according to a feature vector, such as **X,** which shows the essential factors that contribute to the occurrence of a disruptive event. An event belongs to class C with a likelihood computed using the softmax function, where each class has its own parameter vector $\boldsymbol{\theta}_c$.

The softmax function takes linear combinations of inputs and parameters and outputs a collection of probabilities that sum to 1. The chance that an instance belongs to class c is equal to the exponential of the linear score $\theta_c^{\mathrm{T}}\mathbf{X}$, divided by the total of the exponentials of the linear scores for all three classes. This can be written mathematically in Eq. 21, where the projected probabilities are clearly placed in the interval [0,1] by this approach, so that they sum to 1. In this formulation, *exp* denotes the exponential function applied to the term $\theta_c^{\mathrm{T}}\mathbf{X}$.

$$P(y=c|\mathbf{X}) = \frac{\exp\left(\theta_c^{\mathrm{T}}\mathbf{X}\right)}{\sum_{k=1}^{|c|}\exp\left(\theta_c^{\mathrm{T}}\mathbf{X}\right)} \qquad c \in \{low, mid, high\} \tag{21}$$

The proposed approach artificially learns $\boldsymbol{\theta}_c$ during an iterative procedure in training mode, utilizing a maximization of the likelihood function of realized data. This procedure can be done in two ways: minimizing a cross-entropy loss function or maximizing the log-likelihood of the observed data, which enforces the model to place significant importance on the discrepancy between its predicted probabilities and the actual class labels. When the training phase was finished, the model would be able to predict how a new observation falls into each impact level by assigning a probability value to each possible impact level. The model under the maximized likelihood function is given in Eq. (22).

$$\mathscr{L} = -\sum_{i=1}^{N}\sum_{c=1}^{|c|} 1(y_i = c)\log P(y_i = c|\mathbf{x}_i) \tag{22}$$

The function $1(y_i = c)$ returns 1 if observation i belongs to class c, and 0 otherwise. N is the total number of training observations. Furthermore, *log* denotes the logarithm function.

The model's output would be a probability vector where each element expresses the probability that an observation belongs to a relative class. For instance, if the model returns a probability vector [0.25, 0.60, 0.15], it means that the observation with the highest probability would be a medium-impact event. This probabilistic output is beneficial for stochastic programming because it uses a learning-based method to estimate the likelihoods of distinct disruption scenarios and supports judgments about scenario creation, reduction, and planning when uncertainty profoundly affects the system.

To evaluate the proposed logistic regression model's performance, there are several measures in the literature of classification models. In this regard, there are some measures that are frequently used to evaluate a logistic regression model's performance in the literature. We introduce them in the following, and finally, calculate and debate their values in the proposed case study in subsequent sections.

Accuracy: The accuracy measure considers how often predictions are correct across all observations. It is defined mathematically as follows:

$$Accuracy = \frac{\sum_{i=1}^{N} 1(y_i = \widehat{y}_i)}{N}$$

Where y_i is the real label, and $\widehat{y}_i$ is the predicted label. The function $1(y_i = \widehat{y}_i)$ returns 1 if the real label equals the predicted label, and 0 otherwise. Furthermore, N is the number of observations. The favor value for this measure is 1, which means all predictions are correct.

Precision: This measure investigates how many correct positive predictions there are for each class. For example, for class k, we have:

$$Precision_k = \frac{TP_k}{TP_k + FP_k}$$

It can be reported for multi-class by averaging values as macro precision. The favor value for this measure is 1, which means no false positives across any class.

Macro F1 Score: F1 measure combines Precision with Recall, where Recall is defined as follows:

$$Recall_k = \frac{TP_k}{TP_k + FN_k}$$

So, F1 can be formulated for class k as follows:

$$F1_k = 2 \times \frac{Precision_k \times Recall_k}{Precision_k + Recall_k}$$

Now, Macro F1 averages them across all possible classes for identifying a perfect balance of precision and recall for all classes, and its favor value is 1.

$$MacroF1 = \frac{1}{3}\sum_{k=1}^{3} F1_k$$

Macro AUC *(Area Under the ROC Curve)*: This measure is a probability-type index that represents the likelihood that a positive observation is ranked higher than a negative one, where the model predicts the probabilities. The favor value for this measure is 1, which means perfect separability between classes.

Log-loss (Cross-Entropy Loss): The measure reports how the probability distributions of the model's predictions are close to the real labels, where this measure is not limited to getting the right label.

$$Log - loss = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{3} y_{ik}\log(p_{ik})$$

Here, y_{ik} is 1 if sample i belongs to class k, otherwise 0. Also, p_{ik} is the predicted probability of class k. The favor value for this measure is 0, which means perfect confidence and correctness in predictions. Furthermore, *log* denotes the logarithm function.

5. Generalized Benders decomposition solution method

To successfully optimize the suggested mathematical model, choosing an appropriate method for finding the global optimum is crucial. Based on a review by Govindan et al. (2017) Of 79 notable articles on supply chain design for the last decade, Benders decomposition was found to be the most used solution method.

The Generalized Benders Decomposition approach, based on Benders (1962) and Van Slyke and Wets (1969), is a specialized form of Benders decomposition designed for two-stage stochastic programming problems with integer variables in the first stage and continuous non-negative variables in the second stage (Raith et al., 2025). This approach divides the original mixed-integer linear program into two: a master problem (MP) that deals with first-stage decision variables, and one or more Benders sub-problems (BSPs) that deal with optimizing second-stage decisions. To avoid the algorithm getting stuck in a repetitive loop due to static first-stage variables, the dual sub-problem (DSP) is employed instead of the original Benders sub-problem (BSP).

In a general procedure, the algorithm begins by solving the master problem (MP) and obtaining the values for the complex variables used as input parameters for the dual sub-problem (DSP). Solving the DSP provides the values for the other variables. The solutions are then checked for optimality and feasibility. When the problem's solution is infeasible, we need to add a feasibility cut to MP. If the problem's optimality is violated, an optimality cut must be imposed. These two types of cuts are extracted from DSP, and then they are involved in the MP via an iterative procedure to obtain the optimal solution. The MP and DSP formulations for the proposed RSC model are presented below.

Primal sub problem (PSP)

By fixing the first-stage decision variable in the proposed RSC model ($\beta_{jh} = \overline{\beta}_{jh}$), the primal subproblem (PSP) can be formulated. As a result, the PSP would be formulated for each scenario as follows:

$$\begin{aligned}\min Z^{s}_{PSP} = &\sum_{j}\sum_{t}\left[\left(\sum_{r\in r_0}\sum_{i\in i_0} dr_{ij}.q_r.x^{st}_{ijr}\right)+\left(\sum_{r\in r_h}\sum_{i\in i_h} dr_{ij}.q_r.\pi^{st}_{ijrh}\right)\right]+\sum_{j}\sum_{t}\left[\left(\sum_{k\in k_0}\sum_{p\in p_0} dp_{jk}.q_p.y^{st}_{jkp}\right)+\left(\sum_{k\in k_h}\sum_{p\in p_h}\sum_{h} dp_{jk}.q_p.\sigma^{st}_{jkph}\right)\right]\\ &+\sum_{j}\sum_{t}\left[\left(\sum_{p\in p_0} c^{st}_{jp}.v_p\right)+\left(\sum_{p\in p_h}\sum_{h}\rho^{st}_{jph}v_p\right)\right]+\sum_{j}\sum_{t}\left(\sum_{p\in p_h}\sum_{h}\lambda^{st}_{jph}v_p\right)\\ &+\sum_{j}\sum_{t}\left[\left(\sum_{i\in i_0}\sum_{r\in r_0} x^{st}_{ijr}m_r\right)+\left(\sum_{i\in i_h}\sum_{r\in r_h}\sum_{h}\pi^{st}_{ijrh}m_r\right)\right]+\sum_{j}\sum_{t}\left[\left(\sum_{p\in p_0} N^{st}_{jp}.l_p\right)+\left(\sum_{p\in p_h}\sum_{h}\gamma^{st}_{jph}l_p\right)\right]\\ &+\sum_{j}\sum_{t}\left[\left(\sum_{p\in p_0}\chi^{st}_{jp}.\mathscr{H}_p\right)+\left(\sum_{p\in p_h}\sum_{h} w^{st}_{jph}\mathscr{H}_p\right)\right]\end{aligned} \tag{23}$$

$$\sum\nolimits_{k\in k_h}\sum\nolimits_{p\in p_h}\sum\nolimits_{h}\left(\frac{\sum_j\sum_t\sigma^{st}_{jkph}}{\sum_t \mathscr{D}^{st}_{kp}}\right)\geq ep \quad :\theta \tag{24}$$

$$\sum\nolimits_{j} x^{st}_{ijr}\leq \overline{G}_{ir}\mu^{st}_{i} \quad \forall i\in i_0, r\in r_0, t \quad :\theta 1^{st}_{i_0r_0} \tag{25}$$

$$\sum\nolimits_{j}\pi^{s,t+Ld_i}_{ijrh}\leq \overline{G}_{ir}\mu^{st}_{i} \quad \forall i\in i_h, r\in r_h, h, t \quad :\theta 2^{st}_{i_hr_hh} \tag{26}$$

$$c^{st}_{jp}\leq \overline{C}_p \quad \forall j,p\in p_0, t \quad :\theta 3^{st}_{jp_0} \tag{27}$$

$$\rho^{st}_{jph}\leq \overline{C}_p\overline{\beta}_{jh} \quad \forall j,p\in p_h, h, t : \theta 4^{st}_{jp_hh} \tag{28}$$

$$N^{st}_{jp}\leq c^{st}_{jp} \quad \forall j,p\in p_0, t \quad :\theta 5^{st}_{jp_0} \tag{29}$$

$$\gamma^{st}_{jph}\leq \rho^{st}_{jph} \quad \forall j,p\in p_h, h, t \quad :\theta 6^{st}_{jp_hh} \tag{30}$$

$$\chi^{s,t-1}_{jp}+N^{st}_{jp}-\sum\nolimits_{k\in k_0} y^{st}_{jkp}=\chi^{st}_{jp} \quad \forall j,p\in p_0, t \quad :\theta 7^{st}_{jp_0} \tag{31}$$

$$w^{s,t-1}_{jph}+\gamma^{st}_{jph}-\sum\nolimits_{k\in k_h}\sigma^{st}_{jkph}=w^{st}_{jph} \quad \forall j,p\in p_h, h, t \quad :\theta 8^{st}_{jp_hh} \tag{32}$$

$$\frac{\sum_j x^{st}_{ijr}}{\alpha_{rp}}\geq N^{st}_{jp} \quad \forall j,p\in p_0, r\in r_0, t \quad :\theta 9^{st}_{jr_0p_0} \tag{33}$$

$$\frac{\sum_j \pi^{st}_{ijrh}}{\alpha_{rp}}\geq \gamma^{s,t+\eta_j}_{jph} \quad \forall j,p\in p_h, r\in r_h, h, t \quad :\theta 10^{st}_{jr_hp_hh} \tag{34}$$

$$\rho^{st}_{jph}=\rho^{s,t-1}_{jph}+\lambda^{st}_{jph} \quad \forall j,p\in p_h, h, t \quad :\theta 11^{st}_{jp_hh} \tag{35}$$

$$\sum\nolimits_{j} y^{st}_{jkp}\geq \mathscr{D}^{st}_{kp} \quad \forall k\in k_0, p\in p_0, t \quad :\theta 12^{st}_{k_0p_0} \tag{36}$$

$$\sum_j \sigma^{st}_{jkph} \le \mathscr{D}^{st}_{kp} \quad \forall k \in k_h, p \in p_h, h, t \quad : \theta 13^{st}_{k_h p_h h} \tag{37}$$

$$\pi^{st}_{ijrh} \le \mathrm{M}\overline{\beta}_{jh} \quad \forall i \in i_h, j, r \in r_h, h, t \quad : \theta 14^{st}_{i_h j r_h h} \tag{38}$$

Dual sub problem (DSP)

As mentioned earlier, to avoid a cycle due to the first decisions remaining unchanged, Bender's decomposition uses the dual subproblem (DSP). In this regard, we defined the dual variables associated with the PSP constraints, Eqs. (24 to 38), as $\theta, \theta 1, \theta 2$. According to this dual variable definition, the DSP for each scenario s would be formulated as follows:

$$\begin{aligned} \max Z^s_{DSP} = ep.\theta - \sum_{i_0}\sum_{r_0}\sum_t \overline{G}_{ir}\mu^{st}_i \theta 1^{st}_{i_0 r_0} - \sum_{i_h}\sum_{r_h}\sum_h\sum_t \overline{G}_{ir}\mu^{st}_i \theta 2^{st}_{i_h r_h h} - \sum_j\sum_{p_0}\sum_t \overline{C}_p \theta 3^{st}_{jp_0} - \sum_j\sum_{p_h}\sum_h\sum_t \overline{C}_p\overline{\beta}_{jh}\theta 4^{st}_{jp_h h} \\ + \sum_{k_0}\sum_{p_0}\sum_t \mathscr{D}^{st}_{kp}\theta 12^{st}_{k_0 p_0} - \sum_{k_h}\sum_{p_h}\sum_h\sum_t \mathscr{D}^{st}_{kp}\theta 13^{st}_{k_h p_h h} - \sum_{i_h}\sum_j\sum_{r_h}\sum_h\sum_t \mathrm{M}\overline{\beta}_{jh}\theta 14^{st}_{i_h j r_h h} \end{aligned} \tag{39}$$

$$-\theta 1^{st}_{i_0 r_0} + \frac{\theta 9^{st}_{jr_0 p_0}}{\alpha_{rp}} \le m_r + dr_{ij}.q_r \quad \forall i \in i_0, r \in r_0, j, p \in p_0, t \tag{40}$$

$$-\theta 2^{s,t-Ld_i}_{i_h r_h h} + \frac{\theta 10^{st}_{jr_h p_h h}}{\alpha_{rp}} - \theta 14^{st}_{i_h j r_h h} \le m_r + dr_{ij}.q_r \quad \forall i \in i_h, r \in r_h, j, p \in p_h, h, t \tag{41}$$

$$-\theta 3^{st}_{jp_0} + \theta 5^{st}_{jp_0} \le v_p \quad \forall j, p \in p_0, t \tag{42}$$

$$-\theta 4^{st}_{jp_h h} + \theta 6^{st}_{jp_h h} + \theta 11^{st}_{jp_h h} - \theta 11^{s,t+1}_{jp_h h} \le v_p \quad \forall j, p \in p_h, h, t \tag{43}$$

$$-\theta 5^{st}_{jp_0} + \theta 7^{st}_{jp_0} - \sum_{r \in r_0} \theta 9^{st}_{jr_0 p_0} \le l_p \quad \forall j, p \in p_0, t \tag{44}$$

$$-\theta 6^{st}_{jp_h h} + \theta 8^{st}_{jp_h h} - \sum_{r \in r_h} \theta 10^{s,t-\eta_j}_{jr_h p_h h} \le l_p \quad \forall j, p \in p_h, h, t \tag{45}$$

$$-\theta 7^{st}_{jp_0} + \theta 12^{st}_{k_0 p_0} \le dp_{jk}.q_p \quad \forall j, p \in p_0, k \in k_0, t \tag{46}$$

$$-\theta 8^{st}_{jp_h h} + \theta 15^{st}_{k_h p_h h} - \theta 13^{st}_{k_h p_h h} \le dp_{jk}.q_p \quad \forall j, p \in p_h, k \in k_h, h, t \tag{47}$$

$$\theta - \sum_t \mathscr{D}^{st}_{kp}\theta 15^{st}_{k_h p_h h} \le 0 \quad \forall p \in p_h, k \in k_h, h \tag{48}$$

$$-\theta 11^{st}_{jp_h h} \le v_p \quad \forall j, p \in p_h, h, t \tag{49}$$

$$\theta 7^{s,t+1}_{jp_0} - \theta 7^{st}_{jp_0} \le \mathscr{H}_p \quad \forall j, p \in p_0, t \tag{50}$$

$$\theta 8^{s,t+1}_{jp_h h} - \theta 8^{st}_{jp_h h} \le \mathscr{H}_p \quad \forall j, p \in p_h, h, t \tag{51}$$

$$\theta, \theta 1^{st}_{i_0 r_0}, \theta 2^{st}_{i_h r_h h}, \theta 3^{st}_{jp_0}, \theta 4^{st}_{jp_h h}, \theta 5^{st}_{jp_0}, \theta 6^{st}_{jp_h h}, \theta 9^{st}_{jr_0 p_0}, \theta 10^{st}_{jr_h p_h h}, \theta 12^{st}_{k_0 p_0}, \theta 13^{st}_{k_h p_h h}, \theta 14^{st}_{i_h j r_h h}, \theta 15^{st}_{k_h p_h h} \ge 0$$

$\theta 7^{st}_{jp_0}, \theta 8^{st}_{jp_h h}, \theta 11^{st}_{jp_h h}$ Free variables.

Among these, it is necessary to mention that we introduced a dual variable, $\theta 15^{st}_{k_h p_h h}$, for the constraint that was added due to substituting some variables. These substituted variables are employed to simplify Eq. (24), where its original form had a complicated form for formulating the dual form. This substitution was as follows:

$$\frac{\sum_j \sum_t \sigma^{st}_{jkph}}{\sum_t \mathscr{D}^{st}_{kp}} = A^s_{kph}$$

Therefore, Eq. (24) can be reformulated as follows:

$$\sum_j \sum_t \sum_h A^s_{kph} \ge ep$$

Finally, Equations (47) and (48) are formulated by introducing a dual variable corresponding to A^s_{kph}.

Modified dual sub problem (MDSP)

The extreme points cannot be derived when the DSP leads to unbounded, so the proposed algorithm faces interruption in pro-

ceeding with its steps. To address this, a Modified Dual Subproblem (MDSP) is introduced to capture the extreme rays, which can be used in place of extreme points. The MDSP for each scenario s is formulated as follows:

$$\begin{aligned}\max Z^{s}_{MDSP} = ep.\theta - \sum_{i_0}\sum_{r_0}\sum_{t}\overline{G}_{ir}\mu_i^{st}\theta 1^{st}_{i_0r_0} - \sum_{i_h}\sum_{r_h}\sum_{h}\sum_{t}\overline{G}_{ir}\mu_i^{st}\theta 2^{st}_{i_hr_hh} - \sum_{j}\sum_{p_0}\sum_{t}\overline{C}_p\theta 3^{st}_{jp_0} - \sum_{j}\sum_{p_h}\sum_{h}\sum_{t}\overline{C}_p\overline{\beta}_{jh}\theta 4^{st}_{jp_hh} \\ + \sum_{k_0}\sum_{p_0}\sum_{t}\mathscr{D}^{st}_{kp}\theta 12^{st}_{k_0p_0} - \sum_{k_h}\sum_{p_h}\sum_{h}\sum_{t}\mathscr{D}^{st}_{kp}\theta 13^{st}_{k_hp_hh} - \sum_{i_h}\sum_{j}\sum_{r_h}\sum_{h}\sum_{t}M\overline{\beta}_{jh}\theta 14^{st}_{i_hjr_hh}\end{aligned} \tag{52}$$

$$\begin{aligned}ep.\theta - \sum_{i_0}\sum_{r_0}\sum_{t}\overline{G}_{ir}\mu_i^{st}\theta 1^{st}_{i_0r_0} - \sum_{i_h}\sum_{r_h}\sum_{h}\sum_{t}\overline{G}_{ir}\mu_i^{st}\theta 2^{st}_{i_hr_hh} - \sum_{j}\sum_{p_0}\sum_{t}\overline{C}_p\theta 3^{st}_{jp_0} - \sum_{j}\sum_{p_h}\sum_{h}\sum_{t}\overline{C}_p\overline{\beta}_{jh}\theta 4^{st}_{jp_hh} + \sum_{k_0}\sum_{p_0}\sum_{t}\mathscr{D}^{st}_{kp}\theta 12^{st}_{k_0p_0} \\ - \sum_{k_h}\sum_{p_h}\sum_{h}\sum_{t}\mathscr{D}^{st}_{kp}\theta 13^{st}_{k_hp_hh} - \sum_{i_h}\sum_{j}\sum_{r_h}\sum_{h}\sum_{t}M\overline{\beta}_{jh}\theta 14^{st}_{i_hjr_hh} = 1\end{aligned} \tag{53}$$

$$-\theta 1^{st}_{i_0r_0} + \frac{\theta 9^{st}_{jr_0p_0}}{\alpha_{rp}} \le 0 \quad \forall i \in i_0, r \in r_0, j, p \in p_0, t \tag{54}$$

$$-\theta 2^{s,t-Ld_i}_{i_hr_hh} + \frac{\theta 10^{st}_{jr_hp_hh}}{\alpha_{rp}} - \theta 14^{st}_{i_hjr_hh} \le 0 \quad \forall i \in i_h, r \in r_h, j, p \in p_h, h, t \tag{55}$$

$$-\theta 3^{st}_{jp_0} + \theta 5^{st}_{jp_0} \le 0 \quad \forall j, p \in p_0, t \tag{56}$$

$$-\theta 4^{st}_{jp_hh} + \theta 6^{st}_{jp_hh} + \theta 11^{st}_{jp_hh} - \theta 11^{s,t+1}_{jp_hh} \le 0 \quad \forall j, p \in p_h, h, t \tag{57}$$

$$-\theta 5^{st}_{jp_0} + \theta 7^{st}_{jp_0} - \sum_{r\in r_0}\theta 9^{st}_{jr_0p_0} \le 0 \quad \forall j, p \in p_0, t \tag{58}$$

$$-\theta 6^{st}_{jp_hh} + \theta 8^{st}_{jp_hh} - \sum_{r\in r_h}\theta 10^{s,t-\eta_j}_{jr_hp_hh} \le 0 \quad \forall j, p \in p_h, h, t \tag{59}$$

$$-\theta 7^{st}_{jp_0} + \theta 12^{st}_{k_0p_0} \le 0 \quad \forall j, p \in p_0, k \in k_0, t \tag{60}$$

$$-\theta 8^{st}_{jp_hh} + \theta 15^{st}_{k_hp_hh} - \theta 13^{st}_{k_hp_hh} \le 0 \quad \forall j, p \in p_h, k \in k_h, h, t \tag{61}$$

$$\theta - \sum_{t}\mathscr{D}^{st}_{kp}\theta 15^{st}_{k_hp_hh} \le 0 \quad \forall p \in p_h, k \in k_h, h \tag{62}$$

$$-\theta 11^{st}_{jp_hh} \le 0 \quad \forall j, p \in p_h, h, t \tag{63}$$

$$\theta 7^{s,t+1}_{jp_0} - \theta 7^{st}_{jp_0} \le 0 \quad \forall j, p \in p_0, t \tag{64}$$

$$\theta 8^{s,t+1}_{jp_hh} - \theta 8^{st}_{jp_hh} \le 0 \quad \forall j, p \in p_h, h, t \tag{65}$$

$$\theta, \theta 1^{st}_{i_0r_0}, \theta 2^{st}_{i_hr_hh}, \theta 3^{st}_{jp_0}, \theta 4^{st}_{jp_hh}, \theta 5^{st}_{jp_0}, \theta 6^{st}_{jp_hh}, \theta 9^{st}_{jr_0p_0}, \theta 10^{st}_{jr_hp_hh}, \theta 12^{st}_{k_0p_0}, \theta 13^{st}_{k_hp_hh}, \theta 14^{st}_{i_hjr_hh}, \theta 15^{st}_{k_hp_hh} \ge 0$$

$\theta 7^{st}_{jp_0}, \theta 8^{st}_{jp_hh}, \theta 11^{st}_{jp_hh}$ Free variables.

Master problem (MP)

Based on the extreme points and rays derived from the DSP and MDSP, the optimality and feasibility cuts are formulated. Accordingly, the master problem (MP) is constructed as shown in Eqs. (66) to (69), where $\overline{\theta}$ represents the extreme points and $\overline{\overline{\theta}}$ denotes the extreme rays. In this formulation, Eqs. (67) and (68) are referred to as the optimality and feasibility cuts, respectively.

$$\min Z_{MP} = \sum_{j}\sum_{h}u_{jh}\beta_{jh} + \sum_{s}pos_s\varphi_s \tag{66}$$

$$\begin{aligned}\varphi_s \ge ep.\overline{\theta} - \sum_{i_0}\sum_{r_0}\sum_{t}\overline{G}_{ir}\mu_i^{st}\overline{\theta 1}^{st}_{i_0r_0} - \sum_{i_h}\sum_{r_h}\sum_{h}\sum_{t}\overline{G}_{ir}\mu_i^{st}\overline{\theta 2}^{st}_{i_hr_hh} - \sum_{j}\sum_{p_0}\sum_{t}\overline{C}_p\overline{\theta 3}^{st}_{jp_0} - \sum_{j}\sum_{p_h}\sum_{h}\sum_{t}\overline{C}_p\beta_{jh}\overline{\theta 4}^{st}_{jp_hh} \\ + \sum_{k_0}\sum_{p_0}\sum_{t}\mathscr{D}^{st}_{kp}\overline{\theta 12}^{st}_{k_0p_0} - \sum_{k_h}\sum_{p_h}\sum_{h}\sum_{t}\mathscr{D}^{st}_{kp}\overline{\theta 13}^{st}_{k_hp_hh} - \sum_{i_h}\sum_{j}\sum_{r_h}\sum_{h}\sum_{t}M\beta_{jh}\overline{\theta 14}^{st}_{i_hjr_hh}\forall s\end{aligned} \tag{67}$$

$$ep.\bar{\bar{\theta}} - \sum_{i_0}\sum_{r_0}\sum_{t}\overline{G}_{ir}\mu_i^{st}\overline{\overline{\theta 1}}^{st}_{i_0 r_0} - \sum_{i_h}\sum_{r_h}\sum_{h}\sum_{t}\overline{G}_{ir}\mu_i^{st}\overline{\overline{\theta 2}}^{st}_{i_h r_h h} - \sum_{j}\sum_{p_0}\sum_{t}\overline{C}_p\overline{\overline{\theta 3}}^{st}_{jp_0} - \sum_{j}\sum_{p_h}\sum_{h}\sum_{t}\overline{C}_p\beta_{jh}\overline{\overline{\theta 4}}^{st}_{jp_h h} + \sum_{k_0}\sum_{p_0}\sum_{t}\mathscr{D}_{kp}^{st}\overline{\overline{\theta 12}}^{st}_{k_0 p_0}$$
$$- \sum_{k_h}\sum_{p_h}\sum_{h}\sum_{t}\mathscr{D}_{kp}^{st}\overline{\overline{\theta 13}}^{st}_{k_h p_h h} - \sum_{i_h}\sum_{j}\sum_{r_h}\sum_{h}\sum_{t}\mathrm{M}\beta_{jh}\overline{\overline{\theta 14}}^{st}_{i_h j r_h h} \le 0 \forall s \tag{68}$$

$$\sum_{h}\beta_{jh} \le 1 \quad \forall j \tag{69}$$

Since the proposed mathematical model follows a stochastic framework, it is necessary to modify the standard Benders decomposition approach. In this context, a modified Benders decomposition algorithm tailored for stochastic programming (Generalized Benders Decomposition approach) is presented in Algorithm 2.

Algorithm 2. Generalized Benders Decomposition Algorithm

Define $LB \leftarrow -\infty, UB \leftarrow +\infty, iteration \leftarrow 0, UB(s) \leftarrow 0$
while Convergence = yes
  Solve Master Problem (MP)
  $LP \leftarrow Z^*_{MP}$ and save β^*
  For $s \in S$
    Solve Dual Sub Problem (DSP) and get extreme points
    If DSP is unbound
      Solve Modified Dual Sub Problem (MDSP) and get extreme rays
      Generate a feasibility cut in MP by Eq. (68)
    else
      $UB(s) = Z^*_{DSP}$
      Generate an optimality cut of the relevant scenario in MP by Eq. (67)
    End
  End
  If DSP is bounded for all scenarios
    $UB = pos_s \times UB(s) + \sum_j \sum_h u_{jh}\,\beta_{jh}$
  If $UB - LB < threshold$
    Convergence ← yes
  else
    Formulate MP by generating cuts
  End
  $iteration \leftarrow iteration + 1$
End

6. Results and discussion

The results and discussion of the present study are organized into six subsections: (1) the case study overview and data preprocessing stage; (2) learning-based scenario reduction and scenario probability estimation using the proposed learning-based algorithm; (3) configuration of the RSC and sensitivity analysis of key decisions; (4) details of the stochastic solution; (5) evaluation of the solution methodology, including a comparison with commercial solvers; and (6) key managerial insights from the numerical results.

6.1. Case study overview and data preprocessing

The COVID-19 epidemic had a big effect on the car industry all around the world. Not only did the big car companies suffer a lot, but component suppliers did too. These included problems getting raw materials, suppliers not being available, unforeseen staff shortages, and a market that was changing quickly. During this time, the need for cars dropped drastically, while the need for things like surgical masks and face shields rose sharply.

The current study presents a relevant case study investigating an automotive parts SC in Iran (Negarin Sanat Asia). It is related to an assembly company that produces car instrument clusters as a final product on its lines. Assembling each car's instrument cluster requires a pre-manufactured unit and a specific set of wiring components. The COVID-19 pandemic, as a long-term crisis, significantly

affected automotive manufacturers, as car demand declined and, consequently, orders for instrument clusters. Thus, this SC has also profoundly lost its demand during this disruption. Since the COVID-19 pandemic has lasted for an extended period and intensified, SC's decision-makers have felt significant fear of having to close their businesses.

In this regard, the study has developed a repurposable SC network design model to adapt to long-term crises, such as the COVID-19 pandemic, by enhancing production flexibility, where traditional static approaches, such as redundancies, cannot improve the survivability and viability of the SC. When the disruption impact level is low or the condition is non-disruptive, the assembly line works as usual, and the instrument cluster is its main product. At mid or high-impact levels, the SC cannot continue normal operations, so it participates in two emerging product networks, including masks and face shields, that have appeared due to necessity. These emerging products also need specific raw materials. Fabric and elastic are used for mask production, and plexiglass and plastic are utilized for shield production. This participation, which is depicted as integrating emerged SCs, helps SC to ensure its survivability and viability during severe disruptions.

To estimate the proposed problem size, it is necessary to note that the problem involves assembled products, the required raw materials, suppliers, assembly plants, and retailers, which are treated as critical components in both SCs, including the main and emerging. However, the problem encompasses technical elements, such as emerging SCs, scenarios, and planning periods (the time horizon is six periods). As mentioned earlier, the number of scenarios will be determined due to the proposed scenario generation and reduction algorithm outlined in Section 4. The problem's size can be calculated if we consider a representative case for the number of scenarios, such as eight, as follows:

$$|P| \times |R| \times |I| \times |J| \times |K| \times |H| \times |S| \times |T| = 3 \times 6 \times 6 \times 3 \times 6 \times 2 \times 8 \times 6$$

As mentioned earlier, the geographical locations of the suppliers, assembly factories, and customers in this case study define part of the problem. In the main SC, the suppliers are located in Alborz (Negarin Sanat Asia), which provides pre-manufactured units, and Khorasan-Razavi (Sabzevar Cable Car Company), which supplies wiring components. For the emerged SCs, each has two designated suppliers. Behnam Teb Iranian in Tehran gets the fabric for the masks, while Iran Polyester Company in Esfahan gets the elastic.In the shield supply chain, plexiglass is supplied from Tehran (Alborz Plexi Company), and plastic handles from Qazvin (Iran Alborz Company). The locations of assembly sites and retailers are illustrated on the map shown in Fig. 4.

The raw material coefficients required to produce one unit of each product in both the main and emerging SCs are key parameters in configuring an RSC. Accordingly, Table 2 presents the quantities of raw materials needed to assemble each final product.

As mentioned previously, the long-term crisis considered in this study is the COVID-19 pandemic. This disruptive event represents a prolonged crisis that can be temporarily controlled but may resurge over time. These changes in disruption impact levels necessitate the use of new technologies, such as intelligent predictive models, to gain greater transparency into the system state, including the pandemic's intensity and its effects on businesses. To determine the system state, the study assumes three impact levels are imposed due to disruption: low, medium, and high. The low-impact level implies the system faces minimal pandemic effects and continues to operate as usual. The mid-impact level expresses that the pandemic imposes a significant infection rate, which puts the main supply chain under operational limitations. At the high-impact level, the pandemic has a high infection rate, creating extensive limitations on

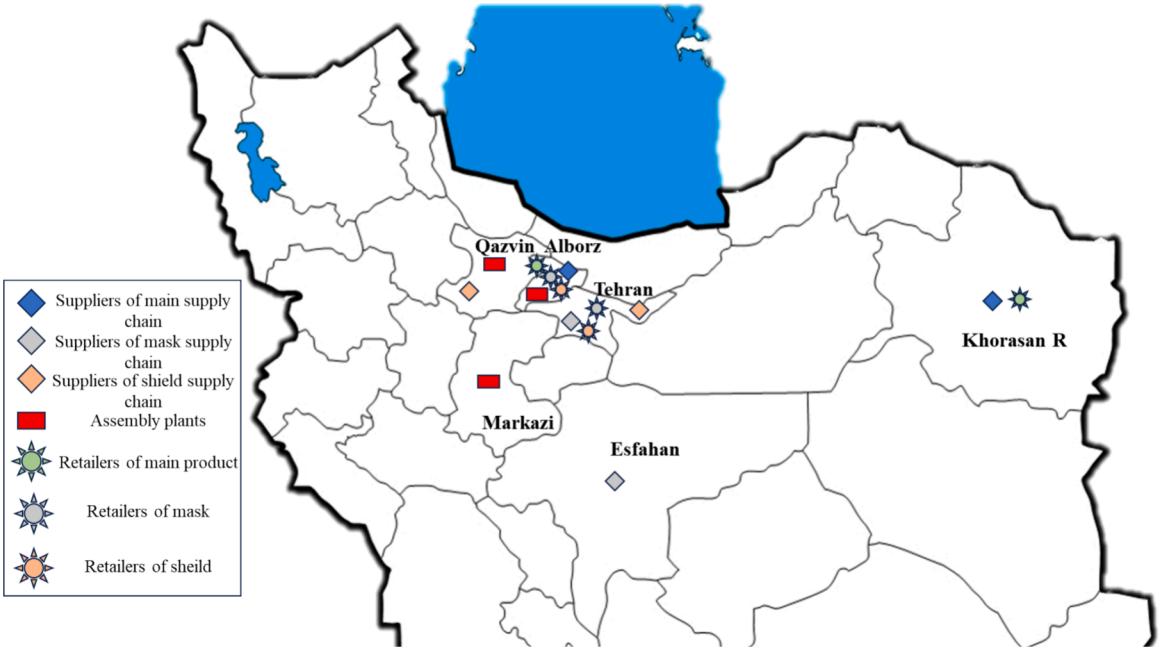


Fig. 4. The locations of assembly sites and retailers.

Table 2
Required raw materials to assemble final products.

Raw material	Instrument cluster (unit)	Mask (unit)	Shield (unit)
Pre-manufactured cluster	1 unit	–	–
Wire	5 m	–	–
Fabric	–	0.1 m^2	–
Elastic	–	0.1 m	–
Plexiglass	–	–	0.07 m^2
Plastic handles	–	–	20 gr

business operations, such as quarantines.

In an RSC, where new SCs may emerge during a long-term crisis, product demand and supplier accessibility must be planned based on the predicted impact level of the disruptive event over the time horizon. Accordingly, the planned demand and supplier availability are determined after constructing the scenario tree and applying the proposed Learning-based scenario reduction method. Table 3 presents the relevant information corresponding to each impact level.

The accessibility matrix for suppliers (μ_i^{st}) and the demand matrix for retailers ($\mathscr{D}_{kp}^{st}$) in the proposed case study in different situations are reported in Table 3, where they will be enabled to determine fitted values based on the construction of the scenario tree and the proposed predictive scenario reduction algorithm in the next section.

As the proposed stochastic programming model follows a learning-based approach, this study utilizes a dataset containing daily records of new COVID-19 cases and deaths, as reported to the WHO (2023). The enabled dataset is related to the COVID-19 pandemic and includes statistics for all countries. Since the case study is on the automotive industry in Iran, it is limited to this country, with records covering the period from January 2021 to December 2022. The features that are reported in this dataset include *new cases, cumulative cases, new deaths, and cumulative deaths*. Since the proposed data-driven model aims for accurate and effective predictions, and the dataset has some feature limitations, additional derived features will be developed on the dataset. The derived features are described in the following section.

The Growth of Cases will be the first new feature introduced to the dataset. It shows the number of new confirmed COVID-19 cases reported on a particular day. This characteristic is based on the difference between the total number of cases on the current day and the previous day.

$$Growth\ of\ Cases = cumulative\ cases^{t} - cumulative\ cases^{t-1}$$

R_0-*Reproduction* rate tells you how many new confirmed cases there are on a given day compared to the number of new cases on the day before. It shows if the epidemic is becoming worse or better from one day to the next.

$$R_0 - Reproduction\ rate = \left(\frac{New\ cases^{t}}{New\ cases^{t-1}}\right)$$

Where R_0 means:

$$\begin{cases} R_0 > 1 & epidemic\ grows \\ R_0 = 1 & steady\ spread \\ R_0 < 1 & outbreak\ declines \end{cases}$$

Growth of Death in this form calculates the number of new deaths reported on a given day by subtracting the cumulative total from the previous day from that of the current day.

$$GrowthofDeath = cumulativedeath^{t} - cumulativedeath^{t-1}$$

The *Death Growth Ratio* compares the number of new deaths on a given day (t) to the number of new deaths on the previous day ($t-1$). It shows how today's death count has changed relative to yesterday'.

$$Death\ Growth\ Ratio = \left(\frac{New\ death^{t}}{New\ death^{t-1}}\right)$$

Table 3
Information on product demand and supplier accessibility associated with each disruption impact level.

Impact	Product Demand (unit)			Supplier Accessibility					
	Instrument cluster	Mask	Shield	Negarin Sanat Asia	Sabzevar Cable Car Company	Behnam Teb Iranian	Iran Polyester Company	Alborz Plexi Company	Iran Alborz Company
Low	5000	–	–	✓	✓	–	–	–	–
Medium	3500	15,000	7000	✓	✓	✓	✓	✓	✓
High	–	25,000	15,000	–	–	✓	✓	✓	✓

These features will be incorporated into the dataset, and the resulting modified dataset will be used in the proposed stochastic phase of the current study.

6.2. Learning-based scenario reduction and scenario probability estimation

To implement the proposed stochastic programming model, the first step involves constructing the scenario tree. As we spoke about before, the COVID-19 pandemic might affect the instrument cluster SC in three ways: low, medium, and high intensity. Based on this assumption, a scenario tree is developed over the defined time horizon, as illustrated in Fig. 5.

According to Fig. 5, the system state can transition to any of the three impact levels in each period. As a result, it is clear that, for the proposed time horizon, the total number of possible scenarios will be calculated via 3^6 combinations of impact levels, which equals 729 scenarios.

According to the constructed scenario tree, the number of possible scenarios in the proposed case study is extremely large, resulting in a complex mathematical model known as an NP-hard problem. In this regard, the proposed predictive scenario reduction and scenario probability estimation algorithm is enabled on the case study.

To efficiently utilize the dataset introduced in the previous section, we defined a response variable called 'pandemic intensity'. This variable represents a categorical value that involves different levels of disruption intensity, such as classes 1, 2, or 3. The assignment of classes to the response variable is as follows:

$$\textit{Pandemic intensity (response variable)} = \begin{cases} 3 & \textit{if Daily infected cases} \geq 30000 \\ 2 & \textit{if } 30000 > \textit{Daily infected cases} \geq 10000 \\ 1 & \textit{if Daily infected cases} < 1000 \end{cases}$$

As mentioned, the proposed model is extended over a six-period time horizon, and the enabled dataset includes data points from the beginning of 2021 to the end of 2022, reported over 24 months. The present study breaks this timeline into six periods, meaning each period spans 4 months in the dataset. Since the proposed predictive algorithm is implemented in Python using the Scikit-learn library, the dataset will be split into two matrices: features and the response variable.

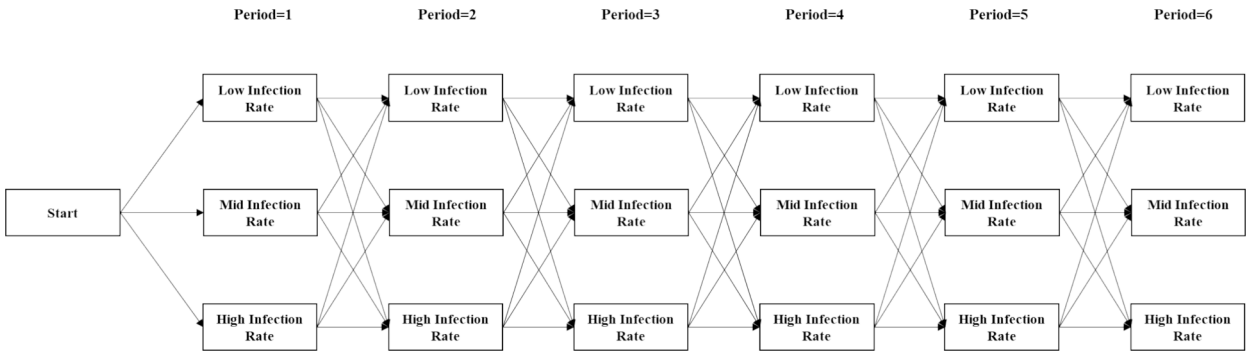


Fig. 5. Scenario tree developed based on the defined time horizon and system states.

Table 4
Learning-based model evaluation.

	Accuracy	Macro F1	Macro AUC	Log-loss	Precision
Iter 1	91.6666	0.8991	1	0.1574	100
Iter 2	83.7837	0.7507	0.9753	0.2252	71.42
Iter 3	94.5945	0.6385	0.997	0.1989	65
Iter 4	88.8888	0.8224	0.9852	0.2125	77.3809
Iter 5	97.1428	0.6927	0.9705	0.0916	70
Iter 6		Dataset included just one class			

Table 5
Probability results for each period and class using the proposed learning-based approach.

		Probability of Each Period Falling into a Disruption Class					
	Periods	t = 1	t = 2	t = 3	t = 4	t = 5	t = 6
Infection Classes	Low Infection Rate	0.73101273	0.101623	0.34452749	0.52426557	0.97769252	1
	Mid Infection Rate	0.26898727	0.48005367	0.53633958	0.32437068	0.02230748	0
	High Infection Rate	0	0.41832332	0.11913293	0.15136376	0	0

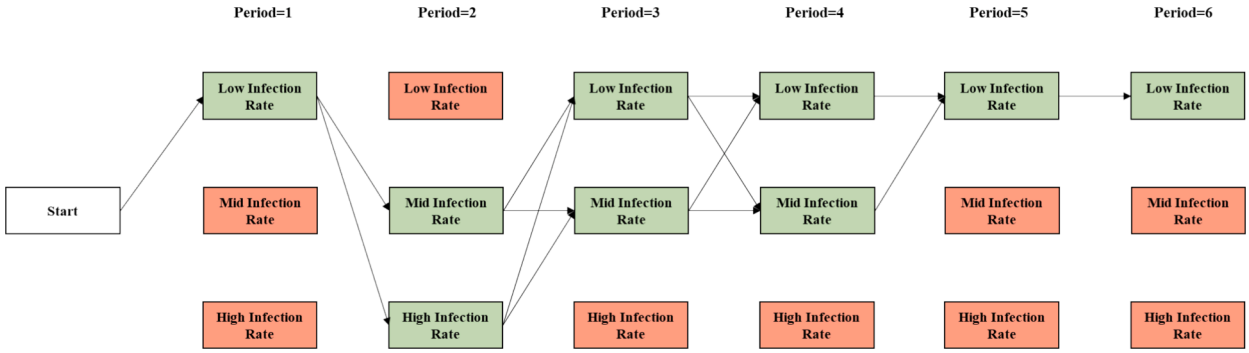


Fig. 6. Refined scenario tree following the proposed learning-based scenario reduction approach.

After the data preprocessing and manipulation performed in the previous section, the processed dataset is used in Algorithm 1 over six iterations, which correspond to the defined periods. Before reporting the algorithm outputs, it is necessary to validate the proposed data-driven model's performance. In this regard, the defined measures, introduced in section 4.2, that evaluate the proposed data-driven model's performance, are calculated under different iterations of the proposed algorithm. Table 4 reports the mentioned measures to evaluate the logistic regression model, which is applied in Algorithm 1, including Accuracy, Precision, Macro F1, Macro AUC, and Log-loss.

According to the accuracy measure, the results indicate that in iterations 1 to 5, the accuracy (all predictions are correct) and precision (no false positives across any class) are more than 50%, which are acceptable values. The next measures, mean Macro F1 and AUC, which interpret a perfect balance of precision and perfect separability between classes, respectively, take their values close to 1, which is their desired value. The final measure is Log-loss, which shows perfect confidence and correctness in predictions, and its desired value is 0. The results indicate Log-loss took values close to 0. The results in Table 4 prove that the learning-based algorithm has a favorable performance in predicting the belonging probability of each class and reducing possible scenarios.

In the first phase of the proposed algorithm, it predicts the probability that each datapoint belongs to each class of the response variable. Then, by averaging predicted probabilities over the relevant section of the dataset in each class, the belonging probability of each class in each period ($\overrightarrow{p}^{t=t}_{c=c}$) is obtained. The resulting probabilities for each period and class are summarized in Table 5.

Since the beginning of the time horizon corresponds to the onset of the pandemic and the end reflects a more controlled situation, the results indicate that the normal mode or low-impact scenarios have the highest probability of occurrence at the start and end. But in other time sections, particularly in the middle periods, the disruption intensity is severe, with a medium-to-high impact level in most periods. However, a high impact level prevails in this part of the time horizon. As is clear from the dataset, this pattern aligns with the expected behavior of the proposed long-term crisis and consistently supports the results.

Once the probabilities of each period falling into a specific disruption class are calculated in the first loop of Algorithm 1, as reported in Table 5, the algorithm proceeds to its second loop. To start the second stage of the proposed algorithm, we set a threshold of 0.3, meaning that if a disruption class has a probability below this threshold for a given period, it will be omitted from the scenario tree. This proposed pruning process reduces the number of possible scenarios meaningfully by eliminating low-probability disruption-impact-level classes. As a result, the initial scenario tree is refined, and the updated, reduced scenario tree is presented in Fig. 6. As mentioned, the threshold was set at 0.3 in this case, where the industrial experts confirmed that the selected scenarios involve all possible situations. However, we have executed a sensitive analysis on different threshold values and reported the number of possible scenarios in each threshold value in Table 6. The results represent the number of remaining scenarios after pruning the scenario tree by increasing the threshold value. All input parameters and model configurations have been defined based on the determined threshold and the pruned scenario tree for executing subsequent experimental analysis.

As mentioned, nodes with predicted probabilities below the proposed threshold have been omitted from the scenario tree, whose final form is illustrated in Fig. 6. This pruning process meaningfully reduces the number of possible scenarios, thereby simplifying the proposed RSC model and strengthening computational tractability. This scenario-reduction process yields the eight scenarios with the highest probability. The reduced scenario tree will shape the proposed model into a structure that is generally more manageable and analytically efficient. The reduced scenarios are outlined in Table 7, which determines the impact level, particularly the infection level in this case, for each period across all scenarios. By assuming independence between observations and then normalizing the probabilities to sum to one, the likelihood of each scenario is calculated using the values from Table 5.

Table 6
Sensitivity of threshold in the learning-based scenario reduction method.

Threshold	0.1	0.15	0.2	0.25	0.3*	0.35
N. Scenario	54	24	16	16	8*	2

* The applied threshold for this case.

Table 7
Details of remaining scenarios by period and infection level, along with corresponding scenario probabilities.

Scenarios	Infection Rate in each period under different scenarios						Normalized Probability
	t = 1	t = 2	t = 3	t = 4	t = 5	t = 6	
s1	Low	Mid	Low	Low	Low	Low	0.129114
s2	Low	Mid	Low	Mid	Low	Low	0.079885
s3	Low	Mid	Mid	Low	Low	Low	0.200997
s4	Low	Mid	Mid	Mid	Low	Low	0.12436
s5	Low	High	Low	Low	Low	Low	0.112511
s6	Low	High	Low	Mid	Low	Low	0.069612
s7	Low	High	Mid	Low	Low	Low	0.175151
s8	Low	High	Mid	Mid	Low	Low	0.108368

Assuming independence, which is enabled in the proposed case study, involves dealing with the selected time horizon for the proposed long-term crises. In this regard, the constructed COVID-19 pandemic dataset shows there is only correlation in a long time period, where the applied time horizon is not long enough in this case. In this regard, the mid-term instabilities dominate the long-term structural dependences. Furthermore, the dataset indicates more noise rather than a trend in the mid-term time horizon that was applied in the proposed case study. Both of the mentioned justifications are another confirmation of the rational utilization of this assumption.

Now, the number of possible scenarios has been significantly reduced, which improves problem tractability and reduces the complexity of the proposed RSC model. Furthermore, the probability of occurrence for each scenario is estimated using the proposed predictive model. These estimated probabilities are subsequently used as inputs for designing the proposed stochastic RSC.

As previously mentioned, two key technical parameters of the RSC, namely $\mathscr{D}_{kp}^{st}$ and μ_i^{st}, are determined based on the disruption intensity in each period as shown in Table 7, along with the corresponding product demand and supplier accessibility for each impact level as detailed in Table 3. These matrices have now been constructed and are presented in Tables 8 and 9.

6.3. Analysis and results of the repurposable supply chain

In this part of the study, we present the results of the proposed RSC model in different aspects. The proposed model is a mixed-integer linear program that was solved using the generalized Benders decomposition method. This method was implemented in GAMS 24.1.2 and operated on a system with a 10th-generation Intel Core i5 CPU and 8 GB of RAM.

As the proposed MILP model is a multi-objective optimization problem addressed using the epsilon-constraint method, we present the resulting Pareto front corresponding to various epsilon values for the second objective. Table 10 reports the CPU time and

Table 8
Supplier accessibility parameter over the time horizon and scenarios.

μ_i^{st}

Supplier.Scenario	Period						**Supplier.Scenario**	Period					
	t1	t2	t3	t4	t5	t6		t1	t2	t3	t4	t5	t6
i01.s1	1	1	1	1	1	1	i01.s5	1	0	1	1	1	1
i02.s1	1	1	1	1	1	1	i02.s5	1	0	1	1	1	1
i1.s1	0	1	0	0	0	0	i1.s5	0	1	0	0	0	0
i2.s1	0	1	0	0	0	0	i2.s5	0	1	0	0	0	0
i3.s1	0	1	0	0	0	0	i3.s5	0	1	0	0	0	0
i4.s1	0	1	0	0	0	0	i4.s5	0	1	0	0	0	0
i01.s2	1	1	1	1	1	1	i01.s6	1	0	1	1	1	1
i02.s2	1	1	1	1	1	1	i02.s6	1	0	1	1	1	1
i1.s2	0	1	0	1	0	0	i1.s6	0	1	0	1	0	0
i2.s2	0	1	0	1	0	0	i2.s6	0	1	0	1	0	0
i3.s2	0	1	0	1	0	0	i3.s6	0	1	0	1	0	0
i4.s2	0	1	0	1	0	0	i4.s6	0	1	0	1	0	0
i01.s3	1	1	1	1	1	1	i01.s7	1	0	1	1	1	1
i02.s3	1	1	1	1	1	1	i02.s7	1	0	1	1	1	1
i1.s3	0	1	1	0	0	0	i1.s7	0	1	1	0	0	0
i2.s3	0	1	1	0	0	0	i2.s7	0	1	1	0	0	0
i3.s3	0	1	1	0	0	0	i3.s7	0	1	1	0	0	0
i4.s3	0	1	1	0	0	0	i4.s7	0	1	1	0	0	0
i01.s4	1	1	1	1	1	1	i01.s8	1	0	1	1	1	1
i02.s4	1	1	1	1	1	1	i02.s8	1	0	1	1	1	1
i1.s4	0	1	1	1	0	0	i1.s8	0	1	1	1	0	0
i2.s4	0	1	1	1	0	0	i2.s8	0	1	1	1	0	0
i3.s4	0	1	1	1	0	0	i3.s8	0	1	1	1	0	0
i4.s4	0	1	1	1	0	0	i4.s8	0	1	1	1	0	0

Table 9
Product demand parameter over the time horizon and scenarios.

Demand for each product under each scenario (Unit)													
Product.Scenario	Period						Product.Scenario	Period					
	t1	t2	t3	t4	t5	t6		t1	t2	t3	t4	t5	t6
p01.s1	5000	3500	5000	5000	5000	5000	p01.s5	5000	0	5000	5000	5000	5000
p1.s1	0	15,000	0	0	0	0	p1.s5	0	25,000	0	0	0	0
p2.s1	0	7000	0	0	0	0	p2.s5	0	15,000	0	0	0	0
p01.s2	5000	3500	5000	3500	5000	5000	p01.s6	5000	0	5000	3500	5000	5000
p1.s2	0	15,000	0	15,000	0	0	p1.s6	0	25,000	0	15,000	0	0
p2.s2	0	7000	0	7000	0	0	p2.s6	0	15,000	0	7000	0	0
p01.s3	5000	3500	3500	5000	5000	5000	p01.s7	5000	0	3500	5000	5000	5000
p1.s3	0	15,000	15,000	0	0	0	p1.s7	0	25,000	15,000	0	0	0
p2.s3	0	7000	7000	0	0	0	p2.s7	0	15,000	7000	0	0	0
p01.s4	5000	3500	3500	3500	5000	5000	p01.s8	5000	0	3500	3500	5000	5000
p1.s4	0	15,000	15,000	15,000	0	0	p1.s8	0	25,000	15,000	15,000	0	0
p2.s4	0	7000	7000	7000	0	0	p2.s8	0	15,000	7000	7000	0	0

Table 10
Resulting Pareto front and associated first-stage decisions in the RSC model across varying epsilon values.

ep	z1 (million Rials)	CPU-time (s)	Optimality Gap	Repurposing for Mask SC			Repurposing for Shield SC			Selected solution
				Alborz	Ghazvin	Markazi	Alborz	Ghazvin	Markazi	
0	518474.74	11.18	0.00	–	–	–	–	–	–	
0.125	585071.81	11	0.00	–	–	–	–	–	✓	
0.251	603606.98	11.183	0.00	–	–	–	–	–	✓	
0.377	627043.21	11.242	0.00	–	–	–	–	–	✓	
0.503	655032.93	11.664	0.00	–	–	–	–	✓	–	
0.628	719837.89	11.481	0.00	–	–	–	–	✓	✓	
0.754	748240.44	11.45	0.00	–	–	–	–	✓	✓	*
0.879	885465.54	11.282	0.00	–	–	–	–	✓	✓	
1.006	928934.30	12.383	0.00	–	–	✓	✓	✓	–	
1.131	979128.40	13.345	0.00	–	–	✓	✓	✓	–	
1.257	1036876.89	15.943	0.00	–	–	✓	✓	✓	–	
	Min-Z2 = 0	Max-z2 = 1.257								

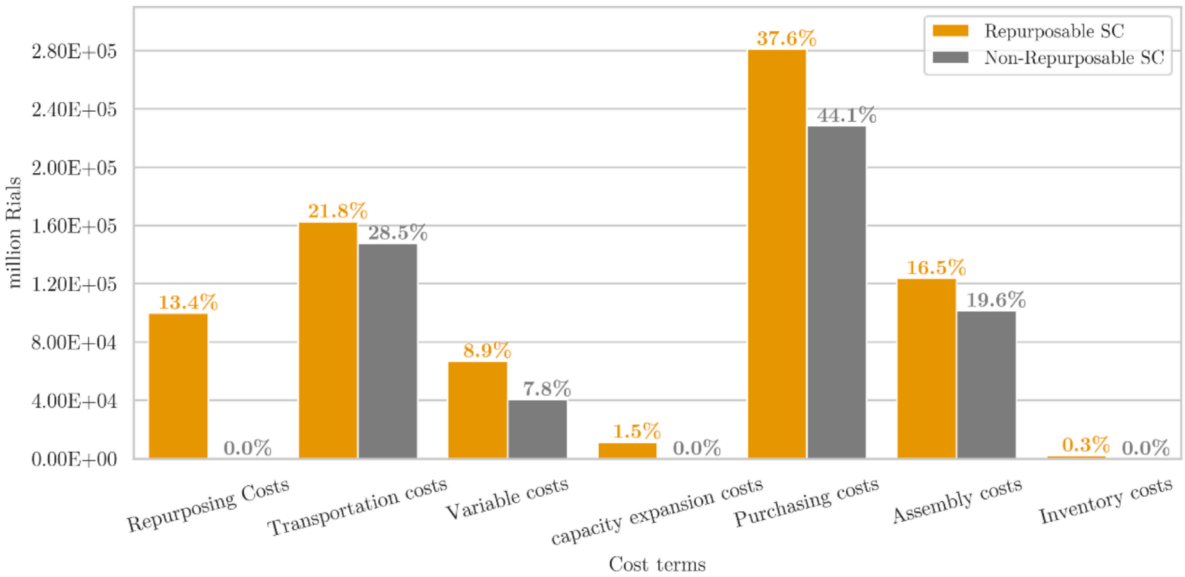


Fig. 7. Proportion of each cost component relative to the total cost within the RSC and non-RSC models.

optimality gap associated with the generalized Benders decomposition method used. The table also highlights the most critical decision in the model, which is the repurposing strategy within the emerging SCs, for each Pareto-optimal solution.

Pareto-optimal solutions represent different aspects of decision-making for SC managers, who can select their preferred solution according to their preferences in a multi-objective optimization model. In this record, Pareto solutions show that by raising the epsilon value, which represents the proportion of fulfilled demand for emerging products over the entire time horizon, the RSC faces more costs. The results are logical because if the decision maker wants to get higher levels of fulfilled demand and income for emerging products, this high-level fulfilment imposes higher costs due to executing more operations. Thus, the decision makers are able to determine their preferred solution in adapting to their preferences for the fulfillment level of emerging products.

Furthermore, the CPU time of the proposed multi-objective problem, which is solved by the generalized Benders decomposition method, has been reported in Table 10. The results indicate, in higher level of fulfilled demand, getting optimal solution takes more time where it means the finding trade-off between economic objective and objective of emerging product fulfilled demand is more difficult in this case. Thus, decision makers have to be more aware of the time consumed for their planning if their favored solution is at this level. Additionally, repurposing decisions in manufacturers in most of the Pareto-solutions are focused on shield SC because it probably creates more economical advantages in relative mask SC. However, when we impose the model to get higher level of fulfilled demand of emerging products, it avoidably involves the mask SC to obtain this trade-off. In conclusion, decision makers have to accept that a higher level of emerging products fulfilling demand will lead to more repurposed manufacturers.

Finally, the selected solution for further analysis is identified. The identified optimal solution yields a cost objective function value of 748,240.44 and a second objective function value of 0.754. In detail, two assembly plants are selected for repurposing, which are located in Ghazvin and Markazi. In both cases, the shield supply chain was preferred over the mask supply chain as an emerging SC involved in the assembly plants.

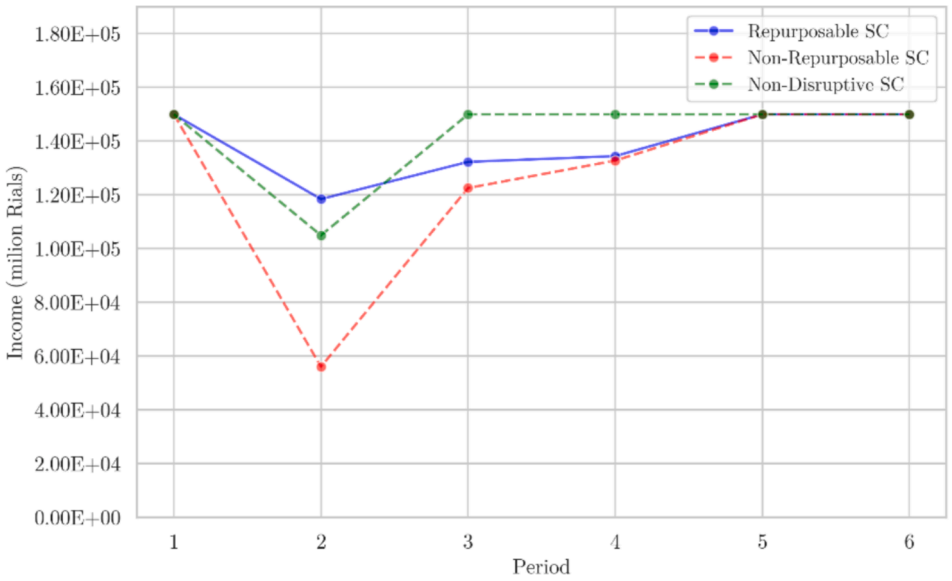


Fig. 8. Supply chain's income over the planning horizon under three conditions: with an RSC, a non-RSC, and in the absence of disruptions.

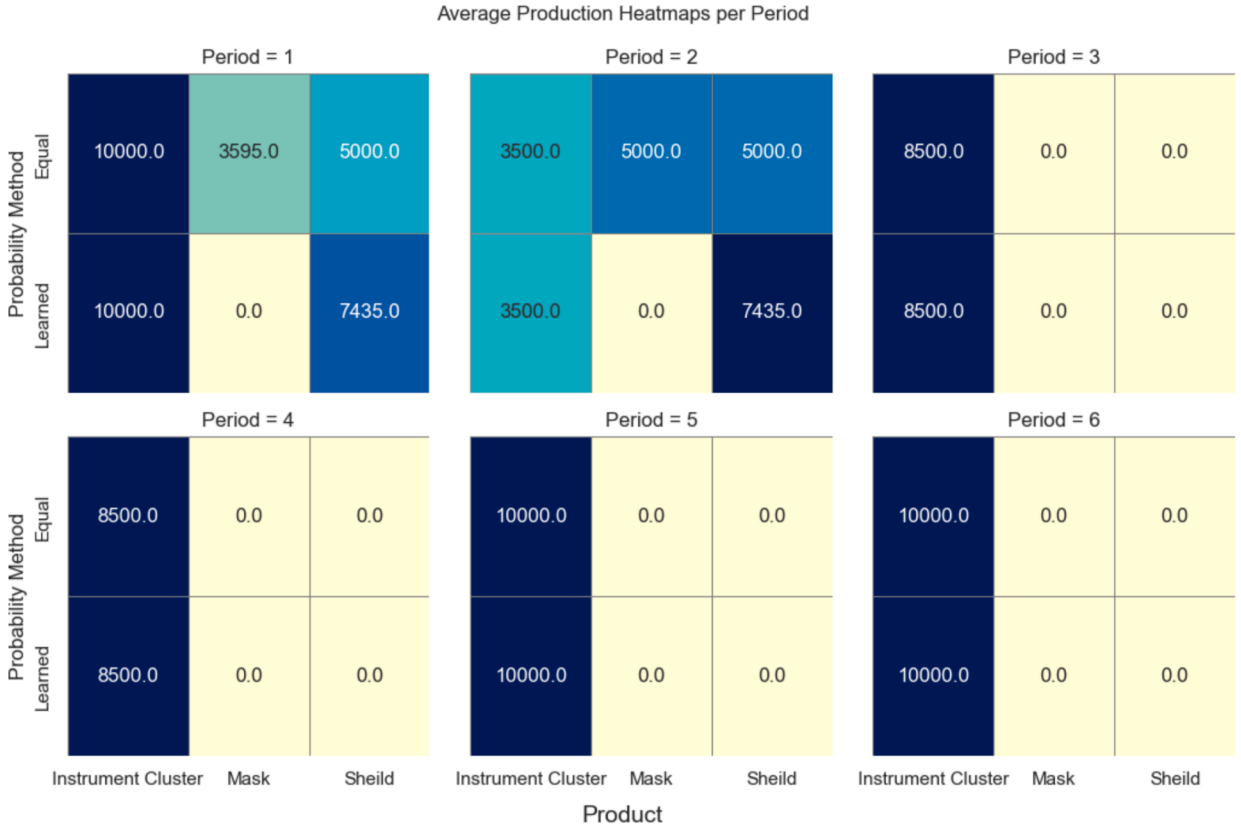


Fig. 9. Average volume of assembled products in both the main and emerging SCs under the assumption of equal probabilities and the proposed approach.

As formulated in section 3, the cost objective function comprises terms such as repurposing, transportation, variable costs, capacity expansion, raw material procurement, assembly operations, and inventory costs. To enhance clarity, we compare the costs of the RSC with those of a non-repurposable counterpart, as illustrated in Fig. 7. Naturally, the non-repurposable network incurs no repurposing costs. Given the RSC's involvement in emerging SCs, its higher overall costs are expected. It is worth noting that three terms have been

affected more than others by repurposing: raw material procurement, transportation, and assembly operations. They show a significant increase over the repurposable system.

The proportion of each cost component relative to the total costs is reported in Fig. 6 for both models, non-repurposable and repurposable, as percentages shown in the bar chart. For instance, for purchasing costs, a significant cost term, the results are reported on the bar chart: 37% and 44.1%, showing the share of total costs in the repurposable and non-repurposable SC models, respectively. The diagram provides insights showing that the purchasing term has the largest share of the cost objective function in both models. At the following levels, assembly and transportation costs are significant components of the overall network costs.

Although the proposed RSC incurs higher costs compared to the non-repurposable model, its involvement in emerging supply chains during long-term disruptions is expected to generate greater income. Fig. 8 illustrates the supply chain's income over the planning horizon under three conditions: with an RSC, a non-repurposable SC, and in the absence of disruptions.

The results in Fig. 8 show that the non-RSC (red line) experienced a sharp decline in income due to disruptions that halted the supply process and assembly line operations. The green line on the diagram shows the SC's income when the disruption does not occur, resulting in uninterrupted operations. The RSC's income, shown by the blue line, demonstrates a notable increase, surpassing even the non-disruptive case in the second period. This indicates that by engaging in emerging supply chains during the peak of disruption, the supply chain was able to generate substantial income (9.7% more than the non-repurposable SC in total). Toward the end of the planning horizon, as the impact of the disruption diminishes, income from emerging SCs declines while revenue from the main SC recovers.

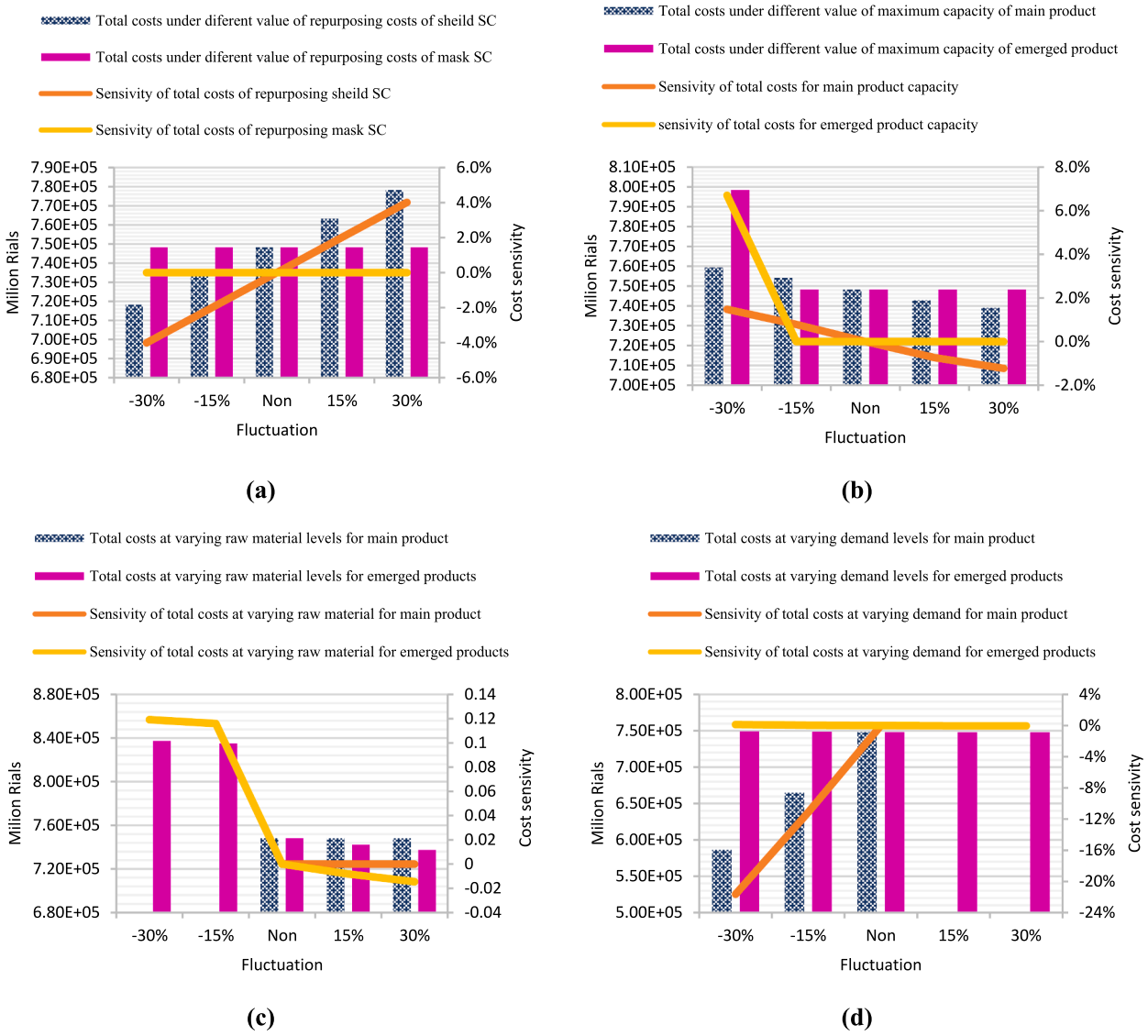


Fig. 10. Sensitivity analysis of key input parameters and their impact on the cost objective function.

As discussed in the previous subsection, the proposed stochastic model incorporates a learning-based method for estimating the probability of different disruption scenarios. To evaluate its impact, we analyze the assembled product volumes over the time horizon under two approaches: one assumes equal probabilities, and the other applies the proposed learning-based probability estimation. Fig. 9 compares the average volume of assembled products in both the main and emerging SCs under these two approaches throughout the planning horizon among different scenarios.

The figure illustrates that the production volume of the main product (instrument cluster) remains consistent across both approaches. However, it reaches its lowest level in period 2 and is reduced in periods 3 and 4. This decline corresponds to the periods of highest disruption impact, particularly in the second period. In contrast, the assembled volumes in the emerging SCs, specifically for masks and shields, show different trends. Under the equal probability approach, the model allocates production to both emerging SCs, influenced by the inclusion of worst-case conditions that actually have very low probabilities. As a result, the model becomes overly conservative. In comparison, the learning-based approach reflects more realistic behavior based on actual data. In this way, the

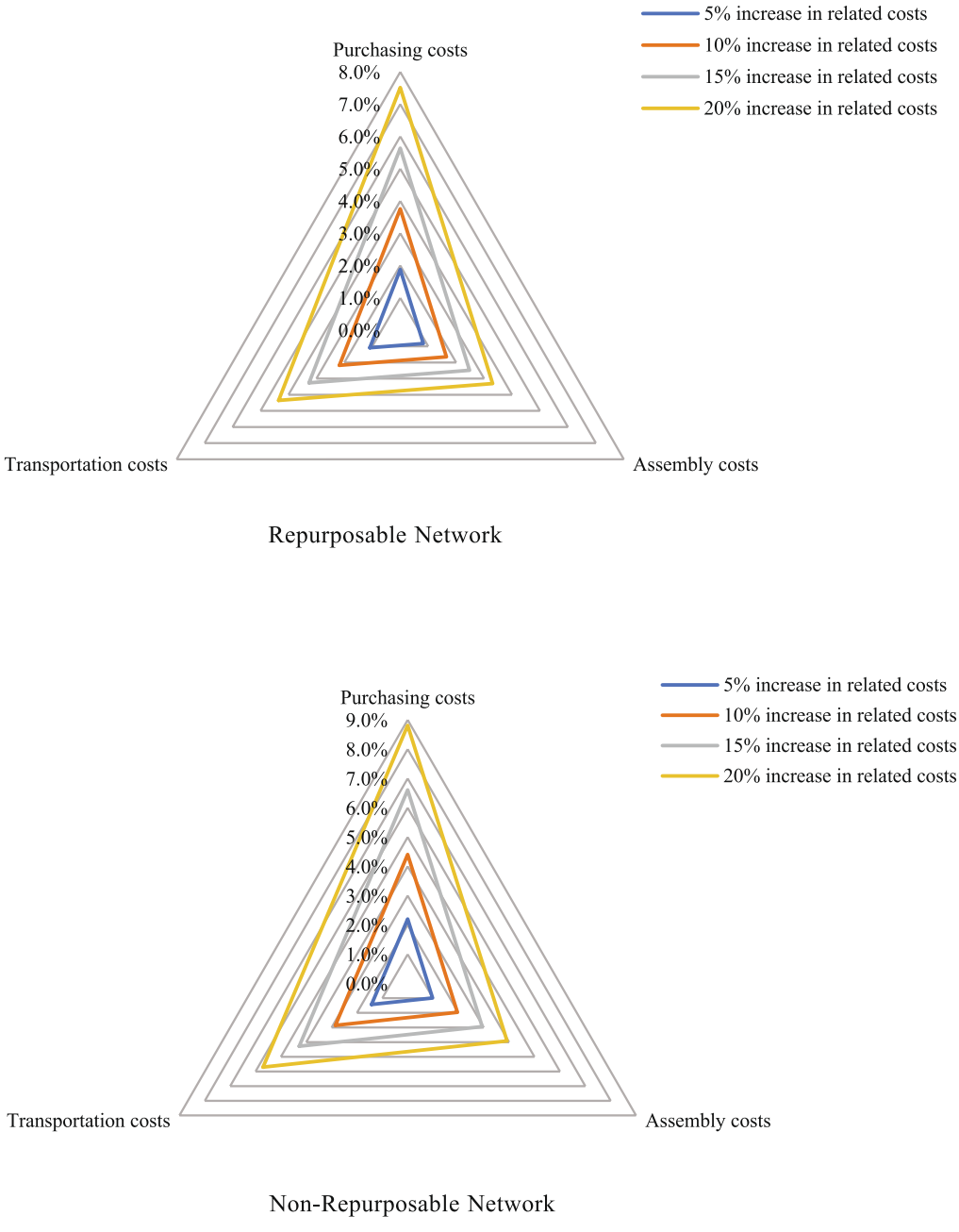


Fig. 11. Sensitivity analysis on cost parameters is conducted for both RSC and non-RSC.

Table 11
Percentage of utilized capacity for each plant under both the main and emerging product SCs.

Assembly Plants			**Alborz**						**Ghazvin**						**Markazi**			
Instrument cluster SC	t1	t2	t3	t4	t5	t6	t1	t2	t3	t4	t5	t6	t1	t2	t3	t4	t5	t6
S1	100	87.5	100	100	100	100	50	0	50	50	50	50	100	87.5	100	100	100	100
S2	100	87.5	100	87.5	100	100	50	0	50	0	50	50	100	87.5	100	87.5	100	100
S3	100	87.5	87.5	100	100	100	50	0	50	50	50	0	100	87.5	87.5	100	100	100
S4	100	87.5	87.5	87.5	100	100	50	0	0	50	50	0	100	87.5	87.5	87.5	100	100
S5	100	0	100	100	100	100	50	0	50	50	50	50	100	0	100	100	100	100
S6	100	0	100	87.5	100	100	50	0	50	0	50	50	100	0	100	87.5	100	100
S7	100	0	87.5	100	100	100	50	0	50	50	50	0	100	0	87.5	100	100	100
S8	100	0	87.5	87.5	100	100	50	0	0	50	50	0	100	0	87.5	87.5	100	100
Sheild SC																		
S1	–	–	–	–	–	–	0	0	0	0	0	0	0	0	0	0	0	0
S2	–	–	–	–	–	–	0	80.15	80.15	80.15	80.15	80.15	0	70	70	70	70	70
S3	–	–	–	–	–	–	80.15	80.15	80.15	80.15	80.15	80.15	70	70	70	70	70	70
S4	–	–	–	–	–	–	80.15	80.15	80.15	80.15	80.15	80.15	70	70	70	70	70	70
S5	–	–	–	–	–	–	0	100	100	100	100	100	0	50.15	50.15	50.15	50.15	50.15
S6	–	–	–	–	–	–	100	100	100	100	100	100	50.15	50.15	50.15	50.15	50.15	50.15
S7	–	–	–	–	–	–	75	75	75	75	75	75	69.33	69.33	69.33	69.33	69.33	69.33
S8	–	–	–	–	–	–	0	0	0	0	0	0	0	0	0	0	0	0

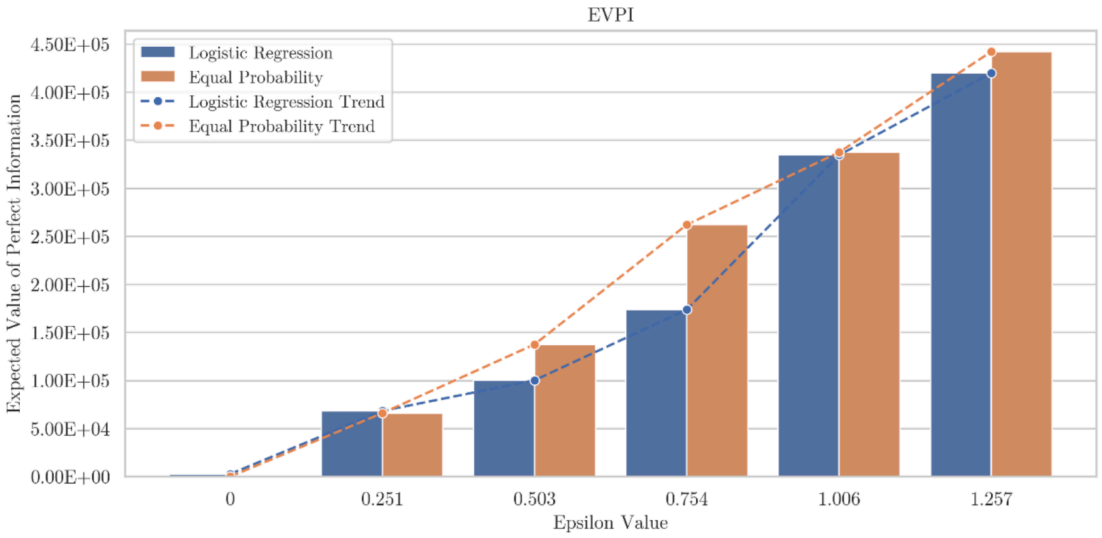


Fig. 12. EVPI computed across epsilon values for learning-based and equal-probability models.

probability of disruption occurrence in each period was accurately estimated using the realized data, which helps avoid unnecessary conservatism imposed on the model by shifting operations, such as repurposing efforts.

In the following, a sensitive analysis will be conducted on several key input parameters, varying these parameters to analyze the behavior of the cost objective function. These parameters include repurposing cost, maximum available capacity at each assembly plant, maximum available raw material from suppliers, and demand levels for both main and emerging products. The outcomes of this sensitivity analysis are presented in Fig. 10, which contains four subfigures.

Fig. 10(a) presents the analysis of repurposing costs. In the optimal solution, the shield supply chain is selected as the emerging supply chain during the long-term crisis. Therefore, increasing the repurposing cost for the shield SC at the assembly plants directly raises the total cost, resulting in a variation ranging from 4 percent below to 4 percent above the initial objective function value. In contrast, changes in the repurposing cost for the mask SC have minimal impact. Raising the cost does not incentivize entry into the mask SC, and even reducing it by up to 30 percent fails to trigger its selection. Fig. 10(b) illustrates the impact of varying the maximum available capacity at each assembly plant. The results indicate that changes in capacity for the main product influence the objective function with a gradual downward trend, meaning that increasing this capacity leads to a reduction in total cost. On the other hand, if we vary the capacity for emerging products from –30% to + 30%, it is clear that a significant reduction occurs first, so the objective value remains flat. This indicates that the expansion of capacity for emerging products only improves efficiency up to a certain level; beyond that, additional capacity does not lead to more economic performance.

Fig. 10(c) shows how changes in the maximum quantity of raw materials available from suppliers affect the total cost. Blank areas

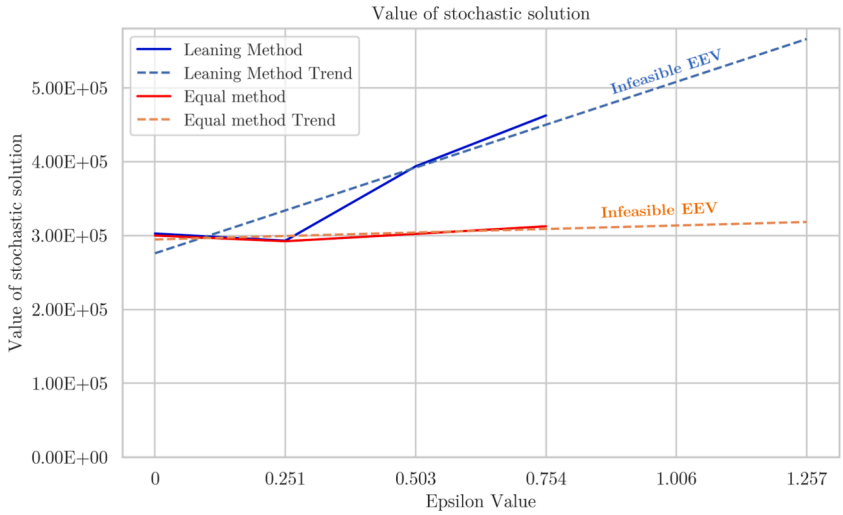


Fig. 13. VSS computed across epsilon values for learning-based and equal-probability models.

in the chart indicate infeasible solutions. For the main product, any reduction in raw material availability renders the model infeasible, making it impossible to meet the mandatory demand. Increasing the availability of raw material more than the demand level, since the objective function is the minimization of expenses, it will be utilized only for meeting the required demand, not more than. For emerging products, lower raw-material limits raise total costs, whereas greater availability produces a modest, tapering cost decline, suggesting that additional raw materials beyond a certain point do not yield further economic benefits. Fig. 10(d) illustrates how variations in demand affect the cost objective function. For the main product, reducing demand to 30% of its original value decreases the total cost to approximately 16% of the initial value. Exceeding the base level in demand will lead to model infeasibility, which may be due to a lack of raw material availability or assembly capacity. On the other hand, varying demand levels for the emerging products within a $\pm$ 30% range do not significantly impact the cost objective function because other model constraints, such as production capacity and raw material availability, will limit the model's ability to achieve higher efficiency.

In this part, a comparative sensitivity analysis will be conducted for both repurposable and non-repurposable networks to compare the supply chain's viability between the two networks. This comparative analysis is illustrated in Fig. 11. In this regard, the relative parameters to the most influential cost terms, including purchasing, transportation, and assembly costs, will fluctuate in a 5% to 20% range to assess their impact on the cost objective function in both networks. As observed, among terms, the purchasing costs show significant sensitivity related to other terms, and, in the next rank, the transportation costs follow it with a moderate effect. The term with the lowest sensitivity and minimal impact on the objective function in the proposed range is Assembly costs.

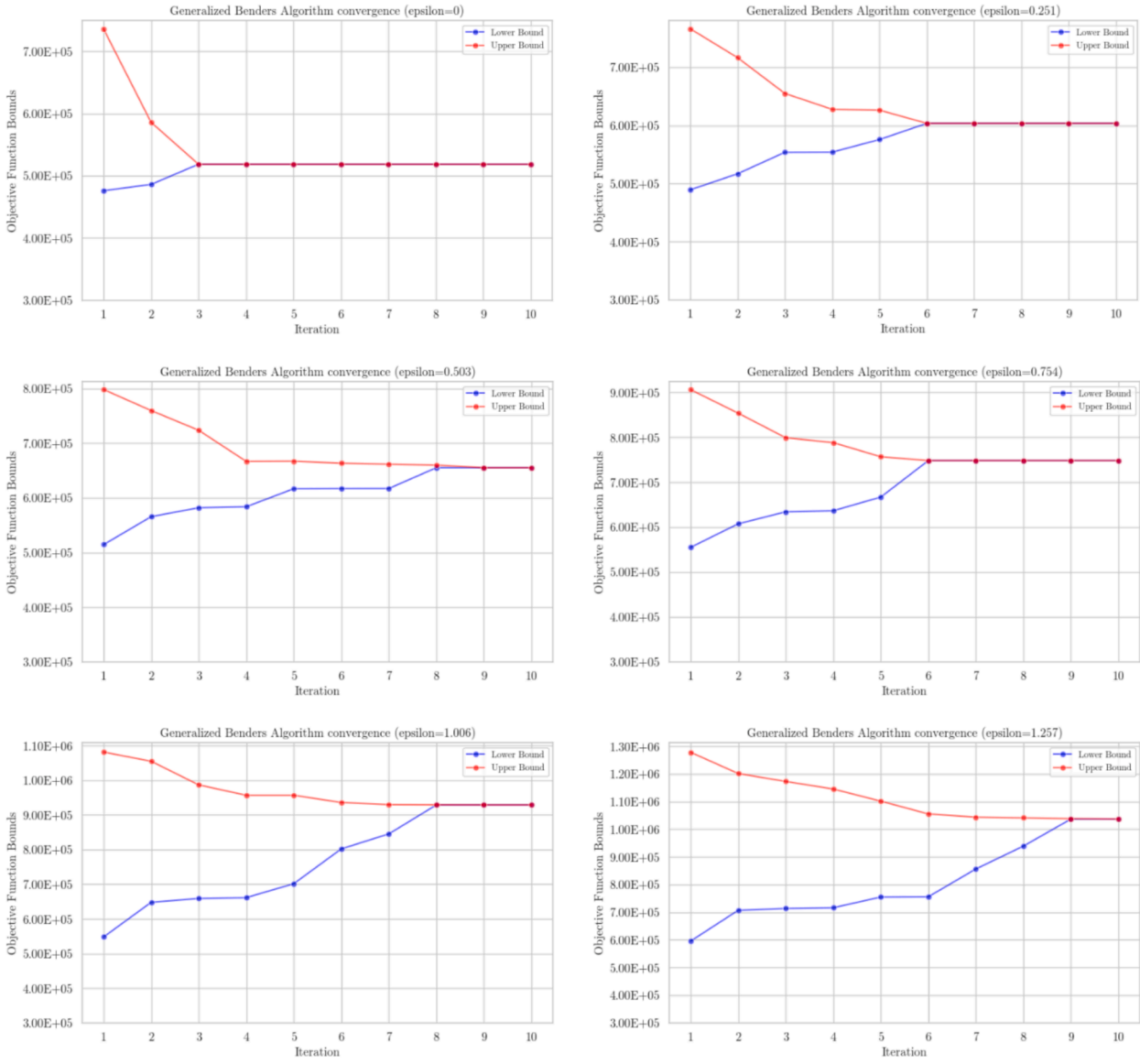


Fig. 14. Convergence behavior of the GBD method applied to the proposed RSC network under varying values of epsilon.

Table 12
Performance comparison: GBD method versus CPLEX solver.

Instance	Problem size	Generalized Benders Algorithm		Basic Benders Algorithm		CPLEX
	$\|P\| \times \|R\| \times \|I\| \times \|J\| \times \|K\| \times \|H\| \times \|S\| \times \|T\|$	CPU-time (s)	Number of Iterations	CPU-time (s)	Number of Iterations	CPU-time (s)
1	$2 \times 3 \times 3 \times 3 \times 3 \times 2 \times 4 \times 3$	7.11	3	5.9	2	0.016
2	$3 \times 4 \times 4 \times 2 \times 4 \times 2 \times 6 \times 4$	9.32	5	7.8	4	2.073
3 (case)	$3 \times 6 \times 6 \times 3 \times 6 \times 2 \times 8 \times 6$	11.45	6	22.78	6	12.016
4	$5 \times 8 \times 8 \times 6 \times 8 \times 4 \times 12 \times 10$	87.49	13	165.3	17	156.4
5	$8 \times 12 \times 12 \times 9 \times 12 \times 6 \times 16 \times 12$	198.21	22	313.1	31	701.38
6	$12 \times 16 \times 16 \times 12 \times 16 \times 10 \times 20 \times 16$	288.55	25	1140.29	39	1005.1
7	$16 \times 20 \times 20 \times 16 \times 20 \times 15 \times 24 \times 20$	511.17	32	2039.21	46	4879.46
8	$20 \times 24 \times 24 \times 20 \times 24 \times 20 \times 30 \times 24$	1023.62	35	8519.5	52	$> 36{,}000$
9	$24 \times 30 \times 30 \times 24 \times 30 \times 24 \times 35 \times 24$	2991.3	39	$> 36{,}000$	–	$> 36{,}000$
10	$30 \times 35 \times 35 \times 30 \times 35 \times 30 \times 35 \times 24$	6088.51	42	$> 36{,}000$	–	$> 36{,}000$

As previously discussed, the main SC includes three assembly plants located in Alborz, Ghazvin, and Markazi. Among these, Ghazvin and Markazi were selected for repurposing in the optimal solution identified on the Pareto front. Table 11 presents the percentage of utilized capacity for each plant under both the main and emerging product SCs.

The results indicate that, within the main SC, the Alborz and Markazi assembly plants have identical capacity utilization levels, whereas the Ghazvin plant operates at no more than 50% of its designed capacity throughout the time horizon and across all scenarios. For the emerging SCs, repurposing has only been selected for the shield SC. Since the Alborz plant was not chosen for repurposing, its capacity utilization for emerging products remains zero. On the other hand, Ghazvin's assembly plant plays a vital role by participating in the shield SC, where capacity utilization ranges from 75% to 100%. This utilization for Markazi's assembly plant falls within the lower range, between 50% and 70%. This highlights that Ghazvin has been repurposed to a greater extent than Markazi. In cases where capacity utilization is zero, the corresponding assembly plant is unable to carry out the assembly process for that SC due to specific periods and conditions that prevent operational continuity, regardless of whether a disruption is present or not.

6.4. Stochastic solution concepts and importance of the learning method

Stochastic programming is widely recognized as a computationally complex and challenging optimization method. To manage this complexity, decision-makers often simplify real-world problems by converting them into deterministic versions. This is usually done by replacing uncertain parameters with their expected values or by solving a deterministic model for each possible realization of uncertainty. However, this approach raises an important question: do such simplifications lead to near-optimal solutions? To answer this question, two important performance measures are introduced. Two important performance measures in stochastic literature include the Expected Value of Perfect Information (EVPI) and the Value of the Stochastic Solution (VSS).

The first measure, EVPI, represents the maximum amount a decision-maker would be willing to pay to obtain complete certainty about future conditions. Assume a set of scenarios models the uncertainty. Here $\boldsymbol{\xi}$ denotes a random variable realized under different cases. The wait-and-see solution (WS) is defined based on the formulation proposed by Madansky (1960) as follows:

$$WS = E_{\xi}\left(\min_{x} z(x, \boldsymbol{\xi})\right)$$

The recourse problem solution (RP) considered in this study is obtained as follows:

$$RP = \min_{x} E_{\xi} z(x, \boldsymbol{\xi})$$

By comparing the wait-and-see solution with the recourse problem solution, the EVPI is defined as the difference between WS and RP, as follows:

$$EVPI = RP - WS$$

After calculating the WS solution, the EVPI is computed under varying epsilon values using the recourse problem solution. The results are presented in Fig. 12 for two cases: the model incorporating the proposed learning-based stochastic programming, and the model assuming equal probabilities across all scenarios. The results show that increasing the epsilon value raises the EVPI. Furthermore, the amount of this rise is more in the model assuming equal probabilities.

This difference indicates that the learning-based model exhibits lower sensitivity to the inherent uncertainty of the problem. This

advantage stems from the use of real data analytics in the proposed learning-based scenario probability estimation method. In contrast, the model assuming equal probabilities is more strongly affected by uncertainty (50.4% higher than the learning-based method at $\varepsilon = 0.754$), suggesting that perfect information would have a higher value in this case.

Generally, solving a deterministic model over all possible scenarios to obtain the WS solution is not feasible in real-world problems. Since acquiring the WS solution requires solving the problem across different scenarios to calculate expectation, it cannot present a single decision as the solution. To address this, we can replace all random variables with their expectations in practice, rather than computing expectations over all scenarios, yielding a specific solution. This solution is known as the Expected Value (EV) problem. it can be formulated as follows:

$$EV = \min_{x} z(x, \bar{\xi})$$

where, $\bar{\xi}$ denotes the expected value of the random variable ξ, and the optimal solution in the EV problem is represented as $\bar{x}(\bar{\xi})$. It is important to note that there is no guarantee that $\bar{x}(\bar{\xi})$ is close to the optimal solution of the recourse problem.

Next, we define the Expected Result of using the EV solution as follows:

$$EEV = E_{\xi}(z(\bar{x}(\bar{\xi}), \xi))$$

EEV reflects how second-stage decisions are optimally determined as a function of $\bar{x}(\bar{\xi})$ and the actual realization of ξ. Finally, the Value of the Stochastic Solution (VSS) is defined as follows:

$$VSS = EEV - RP$$

This concept informs the decision maker of the potential cost incurred by ignoring uncertainty in a stochastic model. Fig. 13 represents a comparative performance in VSS between the learning-based model and the model with equal scenario probabilities across different values of epsilon in the proposed multi-objective formulation. By increasing epsilon, the results indicate that VSS also rises, which represents that ignoring uncertainty leads to greater potential losses. The diagram shows that the EEV becomes infeasible when epsilon exceeds 0.754, so these points have not appeared on the chart.

The observed increase in VSS in the model, assuming equal scenario probabilities, is at a mid-rate significantly lower than that of the learning-based model. This indicates that the learning-based model captures the effects of uncertainty more effectively (48% higher than equal scenario probabilities at $\varepsilon = 0.754$), particularly at higher values of epsilon. In other words, the learning-based model places greater emphasis on accounting for uncertainty compared to the equal-probability approach.

6.5. Assessment of the proposed solution methodology

In this subsection, we evaluate the proposed Generalized Benders Decomposition (GBD) method for solving the RSC network under a long-term disruption. Given the large-scale and complex structure of the model, directly solving it using commercial solvers such as CPLEX proves to be computationally challenging. Therefore, we assess the performance of the proposed solution approach from two perspectives: convergence behavior and comparative efficiency against a commercial solver.

Since the proposed RSC network model is solved using the GBD method, the study examines the algorithm's convergence behavior for different values of epsilon across six subfigures in Fig. 14. As observed, the algorithm has converged rapidly when epsilon is small because, in these cases, the problem becomes more relaxed. As the epsilon value increases, the convergence process requires more iterations. This is due to the DSP facing tighter constraints, which makes it hard to find a solution for improving the bounds' algorithm.

To evaluate the GBD algorithm's performance on the RSC network model, it is helpful to compare a critical solving measure, such as CPU time, with a common commercial solver, such as CPLEX, and the basic Benders decomposition algorithm. In this regard, we provided a comparison of results for 10 instances of the RSC network model across different problem sizes in Table 12, with the parameter set ranging within the values of the case study. As shown, the GBD's performance is relatively slower for instances that have a smaller size (1, 2, and 3) compared to CPLEX. In contrast, it performs significantly better than CPLEX on larger instances (8, 9, and 10). This implies CPLEX fails to obtain an optimal solution within a reasonable computational time on large-scale instances. It is clear that the GBD method solves the RSC large-scale problem in a scalable and effective manner.

Furthermore, it is possible that we compare the proposed GBD method with the basic Benders decomposition algorithm to validate the proposed enhanced approach. In this regard, we formulated the Basic Benders decomposition without generalization for a two-stage stochastic programming model. This formulation has some features that are different from the proposed algorithm, including combining all scenarios into one big deterministic MILP, the subproblem grows huge, and feasibility/optimality cuts won't exploit scenario structure. As mentioned in Table 12, although the basic algorithm has comparative performance in small instances, it loses its performance in large-scale problems. Additionally, in medium-scale problems, it does not have any advantages in CPU time and number of iterations to convergence. In conclusion, we found that the basic algorithm is much slower and leads to losing the benefit of decomposition.

6.6. Managerial implications

The proposed model is applied to real-world data from the instrument cluster assembly supply chain in Iran's automotive industry.

Key managerial insights from the numerical results are summarized below.

Using WHO data from the COVID-19 pandemic, the proposed learning-based algorithm effectively prunes a highly branched scenario tree, induced from the proposed case study, by reducing 729 potential scenarios to the 8 most probable ones for the case study, while also providing predictive probability estimates. Owing to its transparent and data-wise structure, the algorithm is easily adaptable across various industrial contexts.

Although repurposability increases supply chain costs, income analysis reveals that during high-impact periods, the RSC generates 9.7 percent more income than the non-repurposable model in the proposed case. It even surpasses income levels in the non-disruptive case. This demonstrates that repurposability can serve as an effective adaptation strategy for transitioning to the 'new normal' under long-term crises.

The selected optimal solution shows that the shield SC, as an emerging chain during the COVID-19 pandemic, is preferred over the mask supply chain in all selected assembly plants for repurposing. This implies that the shield SC is less costly and more effective in achieving the second objective. The purchasing cost is identified as the most influential cost factor in the RSC. A 20% increase in this cost leads to a 7.5% rise in the objective function, compared to 8.8% in the non-repurposable model in the proposed case study. This confirms that the RSC is more resilient to input cost fluctuations due to its ability to engage with emerging supply chains during long-term disruptions.

By conducting an assembly volume analysis, the study shows that the proposed predictive algorithm indicates less conservatism than the model that assumes equal scenario probabilities in the proposed case study. Less conservatism results from incorporating real-world data into the proposed predictive algorithm, thereby demonstrating realistic behavior in manufacturing.

Stochastic analysis of solutions, using concepts such as EVPI, indicates that the RSC model, equipped with the proposed predictive algorithm, is less sensitive to inherent uncertainty. In contrast, the model assuming equal scenario probabilities exhibits a 50.4% higher sensitivity. This shows that enabling realized data to handle uncertainty effectively reduces the impact of intrinsic uncertainty.

Analysis of the VSS concept shows that the model equipped with the proposed predictive algorithm handles the impact of uncertainty more effectively than the equal-probability model. The results imply that the proposed predictive model integrates decisions with intrinsic uncertainty substantially, with greater emphasis.

Finally, the suggested GBD method demonstrated strong computational performance, addressing large-scale problems within an acceptable CPU time. In contrast, commercial solvers like CPLEX couldn't find the optimal solution in the same situation.

On the other hand, a significant assumption has been enabled in developing the proposed learning-based algorithm to estimate scenario probabilities, which considers the independence of long-term crises' impact levels across the time horizon. Since the mid-term instabilities dominate the long-term structural dependences in the current study, as mentioned in sections 4.1 and 6.2, the correlations are ignorable in the proposed case. In this regard, the managers can ensure that the potential implications of this assumption lead to solutions that are robust enough.

7. Conclusion

In a traditional view, the resilient supply chain involves disruptions that occur for a defined duration, on a predictable scale, and that start the recovery process after the disruption ends. However, the COVID-19 pandemic challenged these assumptions and disrupted supply chains on an unpredictable scale, lasting longer and requiring a recovery process during the disruption itself. This type of disruption redefined what decision-makers believe about the system's disruption configuration. In this regard, returning to the ‘old normal’ was no longer feasible, and the SCs have to adapt to a new normal. To address this, the current study presents a new concept: a repurposable supply chain, in which, under a long-term crisis, the supply chain cannot operate normally and shifts to participate in emerging product supply chains to maintain survivability and viability.

In this regard, the study proposes a MILP model in which a binary variable indicates whether an assembly plant is selected for repurposing to support emerging supply chains. The model is built as a two-stage stochastic formulation to address uncertainty in the impact level of a long-term disruption over the decision-making horizon. Since estimating scenario probabilities and constructing corresponding scenario trees often leads to excessive branching and complexity, digitalization can be a solution. Accordingly, the study introduces a learning-based algorithm that uses machine learning to prune the scenario tree and estimate scenario probabilities using real-time data. Additionally, an exact solution method based on Generalized Benders Decomposition is developed to efficiently solve the large-scale stochastic recourse model.

The model’s results show that the proposed predictive algorithm has significant capability for scenario reduction and probability estimation. Furthermore, income analysis clarifies for managers that access to a new normal is possible in long-term crises by configuring an RSC. Additionally, RSC illustrates more viability and resiliency than non-RSC in relation to cost factors. In regard to intrinsic uncertainty, the enabled realized data plays a pivotal role in reducing uncertainty impact and handling it effectively. Finally, the proposed GBD is able to solve the RSC model in large-scale instances when commercial solvers lose their applicability.

The present study investigates the key aspects of the proposed novel concept in the SC literature, while some limitations remain and open avenues for future research. The following studies can examine the environmental and social implications of RSC to create a sustainable network. Furthermore, it is possible to identify the repurposable concept ability; the researchers conduct a comparative analysis with other adaptive strategies within a viable framework. Additionally, they can consider the long-term crisis that the mid-term instabilities do not dominate the long-term structural dependences and release the independence assumption in the proposed algorithm. This analysis will yield more profound insights into the advantages of the RSC model. On the other hand, the intrinsic uncertainty of impact levels in long-term crises, which affects essentially supplier accessibility and product demand, needs further investigation. In this regard, according to the main supply chain that is critical or not, it can be modeled with a robust formulation with

higher conservatism or softer methods.

CRediT authorship contribution statement

Hani Gilani: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Tom Van Woensel:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

The final formulation of the RSC model, incorporating the ε-constraint method, is presented as follows, where ε is denoted by *ep*.

$$\begin{aligned}\min Z_1^{\varepsilon} = &\sum_j\sum_h u_{jh}\beta_{jh} + \sum_j\sum_s\sum_t Pos_s\left[\left(\sum_{r\in r_0}\sum_{i\in i_0} dr_{ij}.q_r.x_{ijr}^{st}\right) + \left(\sum_{r\in r_h}\sum_{i\in i_h} dr_{ij}.q_r.\pi_{ijrh}^{st}\right)\right] \\ &+ \sum_j\sum_s\sum_t Pos_s\left[\left(\sum_{k\in k_0}\sum_{p\in p_0} dp_{jk}.q_p.y_{jkp}^{st}\right) + \left(\sum_{k\in k_h}\sum_{p\in p_h}\sum_h dp_{jk}.q_p.\sigma_{jkph}^{st}\right)\right] \\ &+ \sum_j\sum_s\sum_t Pos_s\left[\left(\sum_{p\in p_0} c_{jp}^{st}.v_p\right) + \left(\sum_{p\in p_h}\sum_h \rho_{jph}^{st}v_p\right)\right] + \sum_j\sum_s\sum_t Pos_s\left(\sum_{p\in p_h}\sum_h \lambda_{jph}^{st}v_p\right) \\ &+ \sum_j\sum_s\sum_t Pos_s\left[\left(\sum_{i\in i_0}\sum_{r\in r_0} x_{ijr}^{st}m_r\right) + \left(\sum_{i\in i_h}\sum_{r\in r_h}\sum_h \pi_{ijrh}^{st}m_r\right)\right] + \sum_j\sum_s\sum_t Pos_s\left[\left(\sum_{p\in p_0} N_{jp}^{st}.l_p\right) + \left(\sum_{p\in p_h}\sum_h \gamma_{jph}^{st}l_p\right)\right] \\ &+ \sum_j\sum_s\sum_t Pos_s\left[\left(\sum_{p\in p_0} \chi_{jp}^{st}.\mathscr{H}_p\right) + \left(\sum_{p\in p_h}\sum_h w_{jph}^{st}\mathscr{H}_p\right)\right]\end{aligned}$$

$$Z_2 = \sum_{k\in k_h}\sum_{p\in p_h}\sum_h\sum_s Pos_s\left(\frac{\sum_j\sum_t \sigma_{jkph}^{st}}{\sum_t \mathscr{D}_{kp}^{st}}\right) \geq ep$$

$$\sum_j x_{ijr}^{st} \leq \overline{G}_{ir}\mu_i^{st} \quad \forall i \in i_0, r \in r_0, t$$

$$\sum_j \pi_{ijrh}^{s,t+Ld_i} \leq \overline{G}_{ir}\mu_i^{st} \quad \forall i \in i_h, r \in r_h, h, t$$

$$c_{jp}^{st} \leq \overline{C}_p \quad \forall j, p \in p_0, t$$

$$\rho_{jph}^{st} \leq \overline{C}_p\overline{\beta}_{jh} \quad \forall j, p \in p_h, h, t$$

$$N_{jp}^{st} \leq c_{jp}^{st} \quad \forall j, p \in p_0, t$$

$$\gamma_{jph}^{st} \leq \rho_{jph}^{st} \quad \forall j, p \in p_h, h, t$$

$$\chi_{jp}^{s,t-1} + N_{jp}^{st} - \sum_{k\in k_0} y_{jkp}^{st} = \chi_{jp}^{st} \quad \forall j, p \in p_0, t$$

$$w_{jph}^{s,t-1} + \gamma_{jph}^{st} - \sum_{k\in k_h} \sigma_{jkph}^{st} = w_{jph}^{st} \quad \forall j, p \in p_h, h, t$$

$$\frac{\sum_j x_{ijr}^{st}}{\alpha_{rp}} \geq N_{jp}^{st} \quad \forall j, p \in p_0, r \in r_0, t$$

$$\frac{\sum_j \pi_{ijrh}^{st}}{\alpha_{rp}} \geq \gamma_{jph}^{s,t+\eta_j} \quad \forall j, p \in p_h, r \in r_h, h, t$$

$$\rho_{jph}^{st} = \rho_{jph}^{s,t-1} + \lambda_{jph}^{st} \quad \forall j, p \in p_h, h, t$$

$$\sum_j y_{jkp}^{st} \geq \mathscr{D}_{kp}^{st} \quad \forall k \in k_0, p \in p_0, t$$

$$\sum_j \sigma_{jkph}^{st} \leq \mathscr{D}_{kp}^{st} \quad \forall k \in k_h, p \in p_h, h, t$$

$$\pi_{ijrh}^{st} \leq M\overline{\beta}_{jh} \quad \forall i \in i_h, j, r \in r_h, h, t$$

$$\sum_h \beta_{jh} \leq 1 \quad \forall j$$

Data availability

Data will be made available on request.

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